Volatility Forecasting and Value-at-Risk Estimation using ARIMA–GARCH Models and HAR-RV Benchmark

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***Abstract*: Accurate volatility forecasting is fundamental to financial risk management, derivative pricing, portfolio allocation, and regulatory capital assessment under Basel III. This study presents a comprehensive evaluation of ARIMA–GARCH-family models and Heterogeneous Autoregressive (HAR)-type specifications for forecasting daily volatility and estimating market risk in the S&P 500 index from January 2010 to December 2024. The analysis considers GARCH, GJR-GARCH, and EGARCH models under both Gaussian and Student-*t* innovation assumptions, together with HAR-type models constructed from OHLC-based volatility estimators, including Parkinson, Garman–Klass, and Yang–Zhang measures.**

All models are evaluated within a strictly out-of-sample expanding-window forecasting framework. The precision of the volatility forecast is assessed using the mean squared error (MSE), Mean Absolute Error (MAE), and the QLIKE loss function, while the suitability of the risk-model is evaluated through the estimate of the Value-at-Risk (VaR) and Expected Shortfall (ES) combined with the regulatory backtest procedures of Kupiec and Christoffersen.

The empirical results indicate that HAR-type specifications incorporating OHLC-based volatility estimators consistently deliver superior volatility forecasts. These models achieve the lowest forecast losses across multiple evaluation metrics and significantly outperform all benchmark GARCH-family specifications under MSE, MAE, and QLIKE criteria. Pairwise Diebold–Mariano tests further confirm that the forecast improvements are statistically significant, with the results remaining robust after Holm–Bonferroni correction for multiple comparisons. However, asymmetric GARCH models with Student-*t* innovations remain more reliable for practical risk management applications, particularly in tail-risk estimation and regulatory backtesting during periods of elevated market stress such as the COVID-19 crisis and the 2022 bear market.

The findings highlight a fundamental distinction between volatility forecasting accuracy and regulatory risk-model performance. While HAR-type models provide the strongest predictive accuracy for volatility, asymmetric fat-tailed GARCH specifications offer greater robustness for tail-risk measurement and stress-period risk management. These results contribute to the growing literature on volatility modelling by providing a comprehensive comparison of forecasting performance, statistical significance, and regulatory risk adequacy within a unified empirical framework.

Keywords: Volatility forecasting, ARIMA–GARCH models, GJR-GARCH, EGARCH, HAR models, Parkinson volatility, Garman–Klass volatility, Yang–Zhang volatility, Value-at-Risk, Expected Shortfall, Diebold–Mariano test, QLIKE loss, regulatory backtesting, S&P 500.

I. Introduction

The accurate modelling and forecasting of financial volatility constitute central problems in empirical finance, with important implications for derivative pricing, portfolio allocation, and market risk management. Financial returns exhibit well-known stylized facts, most notably *volatility clustering* [1], whereby periods of high volatility tend to be followed by further periods of high volatility. As a result, constant-variance assumptions are generally inadequate for describing financial return dynamics and for measuring risk in real-world markets.

Two broad approaches dominate the volatility forecasting literature. The first is the GARCH family of models, which captures volatility persistence and leverage effects through conditional variance dynamics. The second is the realized-volatility framework, in which volatility is estimated from observed price movements and subsequently modelled using the HAR-RV specification to capture multi-horizon persistence. Prior studies suggest that both approaches can produce accurate volatility forecasts; however, evidence remains mixed regarding their relative performance across different evaluation objectives.

Despite extensive research, three important gaps remain. First, many studies focus exclusively on forecast accuracy while paying limited attention to regulatory risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES). Second, relatively few studies compare asymmetric GARCH specifications and HAR-RV models within a unified expanding-window forecasting framework that avoids look-ahead bias. Third, limited evidence exists regarding the usefulness of OHLC-based realized-volatility estimators, such as the Parkinson, Garman–Klass, and Yang–Zhang measures, when high-frequency data are unavailable.

These limitations are practically important because volatility forecasts are widely used for VaR and ES estimation under modern market-risk frameworks, including Basel III. A model that performs well under conventional forecast-loss metrics may not necessarily provide adequate tail-risk estimates during periods of market stress. Consequently, both forecast accuracy and risk-measure calibration should be evaluated jointly.

This study addresses these issues through a comprehensive comparison of ARIMA-GARCH-family models and HAR-RV specifications using daily S&P 500 data from January 2010 to December 2024. The sample spans multiple market regimes, including the COVID-19 crisis of 2020 and the monetary-policy-driven bear market of 2022, thereby providing a demanding environment for volatility forecasting and risk assessment.

A. Research Objectives

The present study is organized around the following four research questions:

1) Which GARCH-family specification (GARCH, GJR-GARCH, or EGARCH) and conditional error distribution (Gaussian or Student-t) provide the most appropriate representation of volatility dynamics in the S&P 500 return series, as evaluated by information criteria and diagnostic tests?
2) Does the HAR-RV model generate out-of-sample conditional volatility forecasts that are statistically distinguishable from those of the best-fitting GARCH specification, as evaluated under the MSE, MAE, and QLIKE loss functions?
3) Do the parametric VaR estimates produced by each model satisfy the statistical requirements of correct unconditional coverage and violation independence, as assessed by the Kupiec [2] and Christoffersen [3] tests at both the 95% and 99% confidence levels?
4) To what extent does substituting the Gaussian with the Student-t distribution improve VaR calibration accuracy, particularly in the extreme quantiles of the loss distribution?

B. Contributions

This study makes four contributions. First, it compares GARCH-family and HAR-RV models within a unified expanding-window forecasting framework. Second, it evaluates four realized-volatility proxies, including three OHLC-based estimators, within the HAR-RV framework. Third, it extends conventional VaR backtesting to include HAR-RV-implied risk estimates. Fourth, using a fifteen-year sample that spans both the COVID-19 crisis and the 2022 bear market, it provides evidence on the robustness of competing volatility models under stress conditions.

C. Paper Structure

The remainder of this paper is organized as follows. Section II surveys the relevant literature on GARCH-family models, realized volatility, and backtesting frameworks. Section III describes the data and presents summary statistics. Section IV develops the methodological framework. Section V presents the empirical results. Section VI concludes.

II. Literature Review

This section surveys the theoretical and empirical foundations underpinning the models and methods employed in the present study. The review covers four thematic areas: the ARCH/GARCH family of conditional variance models and their error distributions, the HAR-RV framework for realized volatility forecasting, the Value-at-Risk and Expected Shortfall paradigm together with its regulatory context, and the research gap that motivates the current analysis.

A. ARCH/GARCH Models and Their Extensions

The formal econometric treatment of time-varying conditional variance originates with Engle [4], who introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model. Engle demonstrated that the conditional variance of a financial return series can be expressed as a function of lagged squared residuals, providing a rigorous framework for the volatility clustering phenomenon first documented by Mandelbrot [1], whereby large-magnitude returns tend to be followed by further large-magnitude returns regardless of sign.

The principal limitation of the ARCH model, its requirement for a long lag structure to adequately represent volatility persistence, was resolved by Bollerslev [5], who proposed the Generalized ARCH (GARCH) model. By augmenting the variance equation with lagged conditional variances in addition to lagged squared innovations, the GARCH(1,1) specification achieves a parsimonious representation of volatility dynamics that has since become the *de facto* benchmark in the empirical volatility literature. The degree of shock persistence, measured by the sum $\alpha + \beta$, is consistently found to lie close to but below unity for major equity indices, implying mean-reverting yet long-lived volatility [5].

A further empirical regularity that symmetric GARCH does not capture is the leverage effect: negative return shocks generate a disproportionately larger increase in subsequent volatility than positive shocks of equal magnitude. To accommodate this asymmetry, Glosten et al. [6] proposed the GJR-GARCH model, which augments the variance equation with an indicator function that activates only for negative innovations. The associated coefficient γ directly quantifies the leverage effect, and a positive, statistically significant γ constitutes evidence that bad news amplifies variance beyond the symmetric GARCH response. Independently, Nelson [7] introduced the Exponential GARCH (EGARCH) model, which parameterises the logarithm of conditional variance, guaranteeing positivity without explicit parameter constraints and accommodating asymmetric volatility responses through a sign term. Both GJR-GARCH and EGARCH have been shown to produce superior in-sample fits relative to the symmetric GARCH benchmark for major equity indices [6, 7].

B. Error Distributions in GARCH Models

The Gaussian assumption for standardised residuals, while analytically convenient, is widely recognised as inadequate for financial returns. Mandelbrot [1] was among the first to document that the empirical return distribution exhibits heavier tails than the normal distribution implies, meaning that extreme losses occur far more frequently than a Gaussian model would predict. Bollerslev [8] addressed this limitation by proposing the Student-t distribution as the conditional error distribution within the GARCH framework. The degrees-of-freedom parameter ν governs tail thickness and is estimated jointly with the variance parameters. Values of $\nu < 10$ are generally interpreted as evidence of economically meaningful excess tail risk, which translates directly into more conservative and better-calibrated VaR estimates at high confidence levels. Empirical studies consistently confirm that GARCH-t models outperform their Gaussian counterparts in VaR accuracy, especially at the extreme quantiles most relevant for risk management [8].

C. Realized Volatility, OHLC Proxies, and the HAR-RV Model

A methodologically distinct approach to volatility measurement and forecasting has developed around the concept of realized volatility (RV), a nonparametric estimator constructed from high-frequency intraday data. Andersen and Bollerslev [9] established the theoretical foundations of RV as a consistent estimator of the underlying quadratic variation, showing that the sum of squared intraday returns converges to the true integrated variance as the sampling frequency increases.

The principal forecasting model within this tradition is the HAR-RV, introduced by Corsi [10] and grounded in the Heterogeneous Market Hypothesis of Müller et al. [11]. The model regresses next-period realized volatility on its own daily, weekly, and monthly averages, estimated by ordinary least squares. Despite its structural simplicity, this multi-horizon design captures long-memory-like persistence in volatility and achieves competitive out-of-sample forecasting performance relative to more elaborate parametric models [10].

When high-frequency data are unavailable, realized volatility must be approximated using daily-frequency proxies. Four proxies are evaluated in this study. The absolute daily return $|r_t|$, following Ding et al. [12], serves as the close-to-close baseline but is a noisy estimator that discards all intraday price information. The Parkinson [13] range-based estimator is approximately four to five times more efficient than $|r_t|$ under a geometric Brownian motion assumption. The Garman and Klass [14] estimator incorporates the full Open-High-Low-Close information set, yielding further efficiency gains. The Yang and Zhang [15] estimator extends this framework to handle overnight price gaps in a drift-independent manner. Evaluating all four proxies within a unified framework allows the present study to assess both the forecasting gains from richer data inputs and the practical implications of requiring additional price fields beyond closing prices alone.

D. Value-at-Risk, Expected Shortfall, and Backtesting

Value-at-Risk (VaR) quantifies the maximum portfolio loss not exceeded with a given probability over a specified holding period, and has been the dominant market risk measure in regulatory practice since its adoption under the original Basel Accord. Artzner et al. [16] identified a fundamental theoretical shortcoming of VaR: it fails to satisfy the sub-additivity axiom required of coherent risk measures. As a coherent alternative, Artzner et al. proposed Expected Shortfall (ES), which measures the expected loss conditional on the loss exceeding the VaR threshold. The Basel Committee on Banking Supervision formalised the transition from VaR to ES when it adopted ES at the 97.5% confidence level as the primary market risk metric under the Fundamental Review of the Trading Book [17].

The formal validation of VaR through backtesting constitutes both an academic requirement and a binding regulatory obligation. The Proportion of Failures (POF) test of Kupiec [2] provides a likelihood ratio statistic for the null hypothesis that the empirical violation rate equals the nominal confidence level. This limitation was extended by Christoffersen [3], whose Conditional Coverage (CC) framework jointly tests for correct unconditional coverage and independence of violations by modelling the exceedance indicator sequence as a first-order Markov chain. Models that pass the Kupiec test but fail the Christoffersen test are particularly problematic in practice, as they give a false impression of adequate calibration while systematically underestimating risk precisely when it is most consequential.

E. Research Gap and Motivation

The foregoing review establishes that the GARCH family and the HAR-RV constitute the two dominant paradigms for conditional volatility forecasting. The broad consensus in the comparative literature, grounded in the realized volatility framework of Andersen and Bollerslev [9] and the HAR-RV model of Corsi [10], is that HAR-RV outperforms GARCH-family models when high-frequency intraday data are available. However, this consensus rests on three assumptions that the present study explicitly relaxes.

First, existing studies treat realized volatility as a precisely measured quantity constructed from tick-by-tick data, an assumption that does not hold in most practical settings where only daily closing prices are available. While Ding et al. [12] established that absolute daily returns $|r_t|$ carry genuine long-memory information about latent volatility, no systematic study has evaluated whether OHLC-based proxies, which incorporate intraday price range information without requiring tick data, can narrow the performance gap with GARCH under symmetric loss. The question is non-trivial because GARCH filters noise through its recursive variance equation, whereas HAR-RV aggregates noise directly into the regression, and whether this difference favours one paradigm depends on the proxy specification employed.

Second, the comparative literature focuses almost exclusively on symmetric loss functions such as MSE and MAE, neglecting the QLIKE criterion, which penalises variance underprediction more heavily and is better suited to the asymmetric consequences of risk underestimation. More importantly, existing studies do not evaluate whether a model's forecasts satisfy the regulatory tail coverage requirements binding under Basel III [17]. Since point forecast accuracy and VaR validity under the Kupiec and Christoffersen tests need not agree, existing conclusions provide only partial guidance for practitioners who must optimise both objectives simultaneously.

Third, the robustness of both paradigms under qualitatively distinct stress regimes has received insufficient attention. The COVID-19 crash of March 2020, which produced a single-day S&P 500 return of −12.77% (approximately 11.7 unconditional standard deviations), and the sustained bear market of 2022, driven by the most aggressive Federal Reserve tightening cycle in four decades, represent fundamentally different failure modes: the former tests a model's ability to react rapidly to an extreme outlier, while the latter tests its ability to sustain elevated variance forecasts over many months without premature mean reversion. Few existing studies address both within a single expanding-window framework that also covers formal regulatory backtesting requirements.

The present study addresses all three dimensions simultaneously. It evaluates both paradigms across multiple daily-frequency proxy specifications including OHLC-based measures, applies an identical forecast accuracy and regulatory backtesting protocol to both, and does so over a fifteen-year sample encompassing both acute and persistent stress regimes. This design yields conclusions relevant to data-constrained practitioners, regulatory compliance officers, and researchers concerned with model robustness, three audiences whose requirements existing studies address only partially and in isolation.

III. Data

A. Data Source and Description

The empirical analysis is conducted on daily data for the S&P 500 Index (ticker: `^GSPC`), retrieved from Yahoo Finance via the `yfinance` Python library. The primary dataset covers daily closing prices over the period January 5, 2010 to December 30, 2024, yielding a total of 3,772 trading-day observations after removing the first observation lost to differencing. For the OHLC-based proxy analysis (Section IV.E), the full Open-High-Low-Close price series is additionally retrieved, enabling construction of the Parkinson, Garman-Klass, and Yang-Zhang estimators. The S&P 500 is chosen as the empirical subject because it is the most widely followed benchmark for U.S. equity market performance and the canonical test case for volatility models in the academic literature.

The primary variable used throughout the analysis is the continuously compounded daily log return, defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \times 100, \tag{1}$$

where P_t denotes the adjusted closing price on day t and the multiplication by 100 expresses returns in percentage terms. Log returns are employed throughout, as they are time-additive, approximately stationary, and remove the nonstationary trend in raw prices.

Figure 1 provides a visual overview of both series. The upper panel displays the index level over the full sample, growing from approximately 1,100 in January 2010 to nearly 6,000 by end-2024, with the two stress episodes highlighted in shading. The lower panel plots daily log returns, making the key empirical regularities immediately visible: the return series fluctuates around zero with no discernible trend, but its dispersion is clearly not constant over time. Long stretches of relatively small returns are punctuated by brief episodes of extreme volatility, most visibly in the COVID-19 crash of March 2020 and the elevated turbulence of 2022. This pattern of *volatility clustering* is the primary empirical motivation for the GARCH family of models.

The fifteen-year sample encompasses two qualitatively distinct stress episodes: the COVID-19-induced dislocation of early 2020, which produced a single-day return of −12.77% (approximately 11.7 unconditional standard deviations), and the Federal Reserve rate-hiking cycle of 2022, which generated elevated and persistent volatility over several months. Together, these episodes provide a demanding out-of-sample evaluation environment for the candidate models (see Appendix Table 23 for a full regime chronology).

B. Descriptive Statistics

Table 1 reports the summary statistics for the full sample of log returns. Several features of the distribution are immediately apparent and are directly relevant to the model selection decisions made in subsequent sections.

Table 1 Descriptive Statistics of S&P 500 Daily Log Returns (January 2010 – December 2024). Returns are expressed in percentage terms. The Jarque-Bera statistic tests the null hypothesis of normality.

Statistic	Value
Observations	3,772
Mean (%)	0.0438
Standard Deviation (%)	1.0890
Minimum (%)	−12.7652
25th Percentile (%)	−0.3810
Median (%)	0.0681
75th Percentile (%)	0.5675
Maximum (%)	8.9683
Skewness	−0.7262
Excess Kurtosis	13.2144
Jarque-Bera Statistic	27,696.63
Jarque-Bera p-value	< 0.001

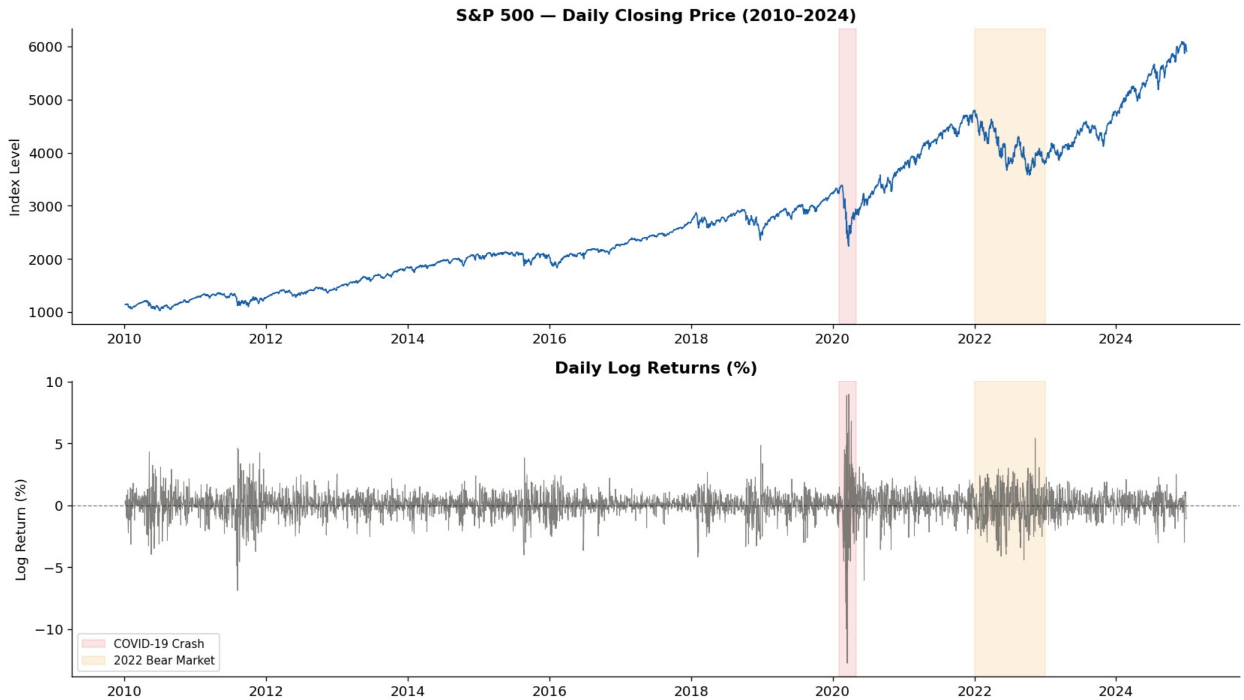

Fig. 1 S&P 500 Daily Closing Price and Log Returns (January 2010 – December 2024). The upper panel shows the index level; the lower panel shows daily log returns (in %). Pink shading indicates the COVID-19 crash period (Q1 2020); orange shading indicates the 2022 bear market. Volatility clustering is clearly visible in the lower panel: large-magnitude returns concentrate in the two shaded episodes, while the majority of the sample exhibits comparatively low dispersion.

The mean daily log return of 0.0438% is close to zero, consistent with the near-martingale behavior expected of daily equity returns. The standard deviation of 1.0890% represents the unconditional, average level of daily risk, but as discussed below, this figure conceals substantial time-variation in the second moment.

The negative skewness of −0.7262 indicates that the left tail of the return distribution is heavier than the right tail: large losses occur more frequently than gains of equal magnitude. This asymmetry is one of the best-documented stylized facts of equity returns and has a natural interpretation in terms of the leverage effect, whereby falling equity prices raise a firm's financial leverage and thereby amplify subsequent return volatility.

The excess kurtosis of 13.2144 is the most diagnostic statistic for model selection purposes. A value this large – more than four times the kurtosis of a normal distribution – implies that extreme return realizations, both positive and negative, occur far more frequently than a Gaussian model would assign non-negligible probability. This is precisely the *fat-tail* property documented by Mandelbrot [1]. In practical terms, it means that the largest losses in the sample, such as the −12.77% daily return recorded during the COVID-19 crash of March 2020, would be assigned a probability of effectively zero under the Gaussian assumption. The Jarque-Bera statistic of 27,696.63 (with $p < 0.001$) formally rejects the null hypothesis of normality at any conventional significance level. Together, these findings provide a strong a priori justification for employing the Student-t distribution rather than the Gaussian assumption in the GARCH variance equations, and they motivate the comparative evaluation of both distributional assumptions that is central to the present study.

Figure 2 provides a visual confirmation of these departures from normality. The left panel superimposes a fitted Gaussian density on the histogram of log returns: the empirical distribution is markedly more peaked at the center (leptokurtic) and has noticeably heavier tails on both sides, with several observations extending well beyond ±5% that would be essentially impossible under the fitted normal curve. The right panel presents a normal quantile-quantile (QQ) plot, in which perfect normality would produce a straight line. The pronounced S-shape deviation confirms heavy tails in both directions: the most extreme negative returns (bottom-left) and positive returns (top-right) lie far above and below the reference line, respectively. This S-shape is the graphical signature of a leptokurtic distribution and provides further visual support for the use of fat-tailed conditional distributions in the GARCH specifications.

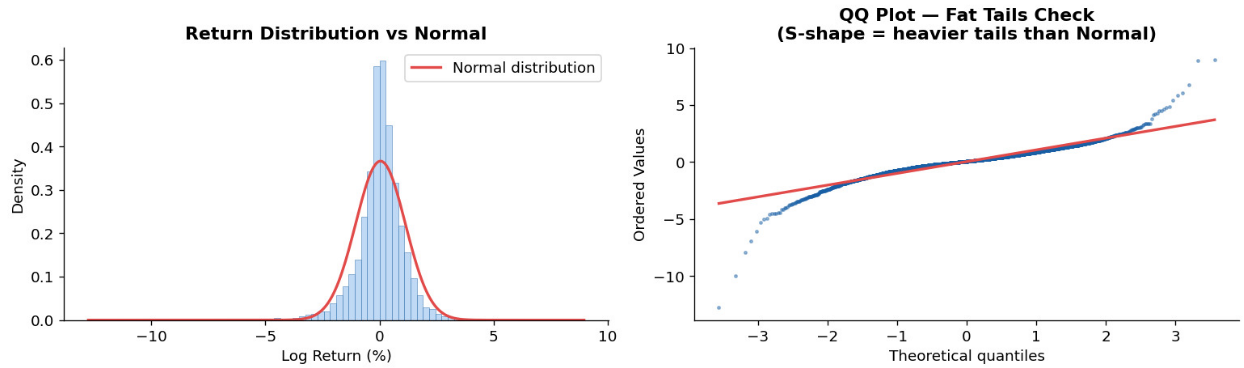

Fig. 2 Return Distribution and Normal QQ Plot for S&P 500 Daily Log Returns (January 2010 – December 2024). ***Left panel:*** **Histogram of log returns (blue bars) with a fitted Normal density overlay (red curve). The empirical distribution is more peaked and has heavier tails than the Normal, particularly on the left side.** ***Right panel:*** **Normal QQ plot. Deviations from the 45-degree reference line (red) indicate departures from normality. The characteristic S-shape confirms fat tails in both the lower and upper extremes of the distribution. Skewness** $= -0.7262$**; Excess Kurtosis** $= 13.2144$**; Jarque-Bera** $p < 0.001$**.**

C. Train/Test Split

The dataset is partitioned into a training set and a test set following a strictly chronological split, with no shuffling of observations. This approach is essential for time series data: any shuffling or random sampling would introduce look-ahead bias by allowing the model access to future information during estimation, which would produce artificially optimistic out-of-sample performance statistics.

An 80/20 split is applied, yielding a training set of 3,017 observations covering January 5, 2010 to December 28, 2021, and a test set of 755 observations covering December 29, 2021 to December 30, 2024. The training set is used for in-sample parameter estimation and model selection via the Akaike and Bayesian information criteria. The test set serves as the out-of-sample evaluation window for volatility forecast accuracy, VaR calibration, and formal backtesting. All forecast evaluation is conducted using an expanding-window scheme, described in detail in Section V, to ensure that each one-step-ahead forecast uses only information available up to the forecast origin.

D. Major Market Events in the Sample

The fifteen-year sample period contains a rich cross-section of market regimes that provides a demanding test of the candidate models. Table 23 in Appendix summarizes the principal episodes.

The COVID-19 episode is particularly important for the evaluation of VaR models. The daily return of -12.77% recorded on March 16, 2020 represents a realization that is approximately 11.7 standard deviations below the sample mean under the Gaussian assumption, an event with a theoretical probability of effectively zero under normality. This single observation, along with the cluster of large negative returns in March 2020, is a direct empirical demonstration of why fat-tailed conditional distributions are necessary for credible risk measurement.

The 2022 bear market, by contrast, was a slower-moving but persistent stress episode driven by the Federal Reserve's most aggressive rate-hiking cycle in four decades. Unlike the sharp V-shaped shock of 2020, the volatility of 2022 was elevated and sustained over several months, which tests the ability of each model to capture persistent volatility regimes through its variance dynamics. Together, these two episodes represent qualitatively different types of market stress and thereby provide a comprehensive evaluation environment for assessing the robustness of the candidate volatility models and their implied VaR estimates.

IV. Methodology

This section presents the complete modelling pipeline employed in the present study. The modelling pipeline proceeds through eight stages: (1) log-return construction and stationarity testing, (2) ARIMA mean equation specification, (3) ARCH-LM diagnostic testing, (4) estimation of six GARCH-family models, (5) construction of four realized volatility proxies from OHLC data, (6) HAR-RV estimation across all proxy variants, (7) expanding-window out-of-sample forecasting, and (8) VaR/ES computation together with formal backtesting. Figure 3 summarizes this workflow.

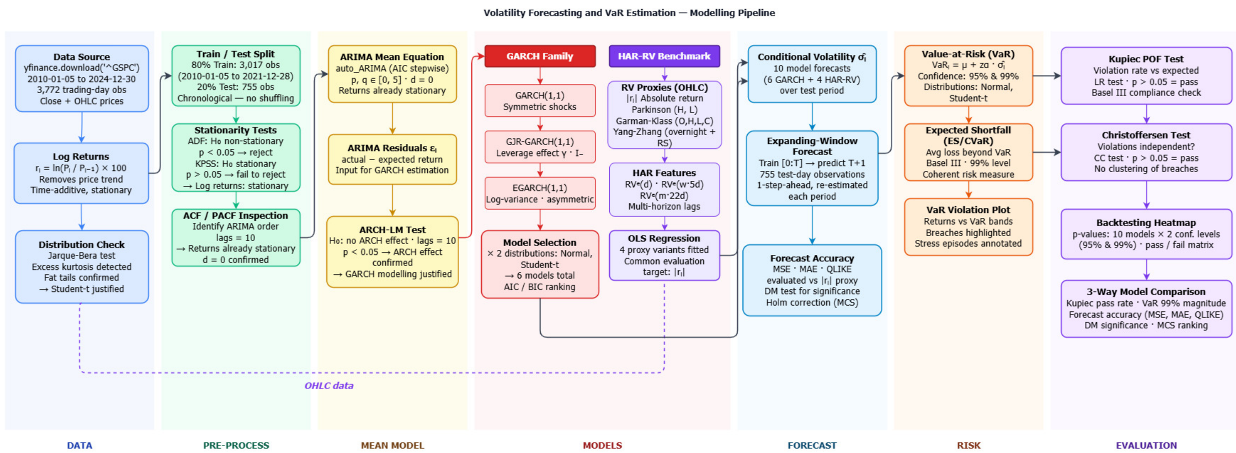

Fig. 3 Workflow pipeline for the present study.

A. Log Returns and Stationarity Testing

As established in Section II, the primary modelling variable is the daily log return $r_t = \ln(P_t/P_{t-1}) \times 100$. Before fitting any time-series model it is necessary to verify that the return series is stationary, meaning that its mean, variance, and autocovariance structure do not change over time. A non-stationary series would invalidate the asymptotic properties of all subsequent estimators.

Two complementary tests are applied, chosen because they test opposite null hypotheses and therefore provide a more robust joint assessment than either test alone.

The **Augmented Dickey-Fuller (ADF)** test [18] has the null hypothesis H_0: the series contains a unit root (non-stationary). The test statistic is:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{k} \delta_i \Delta y_{t-i} + \varepsilon_t, \tag{2}$$

where $\gamma = 0$ under the null. Rejecting H_0 (i.e. $p < 0.05$) provides evidence of stationarity.

The **KPSS** test [19] has the null hypothesis H_0: the series is stationary. Failure to reject H_0 (i.e. $p \geq 0.05$) is consistent with stationarity. Together, rejecting ADF and failing to reject KPSS constitutes the strongest possible combined evidence that the series is $I(0)$ stationary.

B. ARIMA Mean Equation

Volatility models such as GARCH are applied to the residuals of a mean equation rather than to raw returns directly. This ensures that any predictable linear structure in the conditional mean is removed first, so that the GARCH model captures only the dynamics of the conditional variance. The mean equation takes the general ARIMA(p, d, q) form:

$$r_t = \mu + \sum_{i=1}^{p} \phi_i r_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t, \qquad \varepsilon_t \sim (0, \sigma_t^2), \tag{3}$$

where μ is the unconditional mean, ϕ_i are the autoregressive (AR) coefficients governing the dependence of r_t on its own past values, θ_j are the moving-average (MA) coefficients governing the dependence on past innovations, and ε_t is the mean-zero innovation sequence with time-varying conditional variance σ_t^2. Since log returns are already $I(0)$ stationary (as confirmed by the stationarity tests), the integration order is fixed at $d = 0$ throughout.

The optimal lag orders (p, q) are selected via a stepwise search over the AIC criterion using the `auto_arima` function from the `pmdarima` library, with maximum search bounds of $p, q \leq 5$. The AR coefficients ϕ_i and MA coefficients θ_j capture any weak autocorrelation in the return series, and the resulting residuals $\hat{\varepsilon}_t$ constitute the input to the GARCH variance models.

Figure 4 displays the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the training-set log returns up to 40 lags. The interpretation of these plots follows a standard diagnostic rule: a sharp cut-off in the PACF after lag p suggests an AR(p) process, while a sharp cut-off in the ACF after lag q suggests an MA(q) process. In both panels, virtually all bars lie within the 95% confidence band (shaded region), indicating that the autocorrelation in the level of returns is economically negligible. The absence of strong linear structure confirms that daily log returns are very close to serially uncorrelated, which is consistent with the efficient market hypothesis at the daily frequency. The `auto_arima` stepwise search, which formally minimizes AIC, selected ARIMA(4, 0, 0) as the best-fitting mean specification.

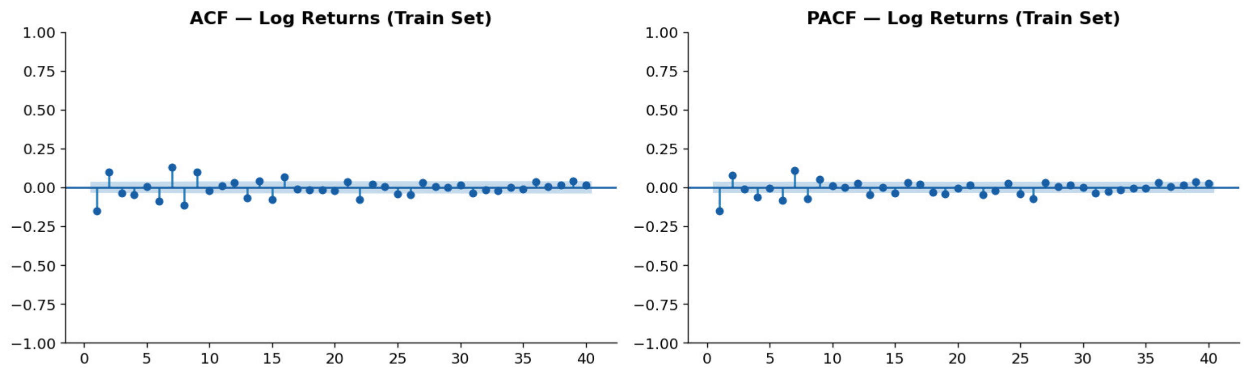

Fig. 4 Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of S&P 500 daily log returns on the training set (January 2010 – December 2021, T = 3,017 observations). The shaded blue band represents the 95% confidence interval under the null hypothesis of zero autocorrelation. *Left panel:* ACF up to 40 lags. *Right panel:* PACF up to 40 lags. Nearly all bars lie within the confidence band, indicating that log returns exhibit negligible linear serial dependence. The selected ARIMA order is (4, 0, 0).

C. ARCH Effect Test

Before fitting a GARCH model, it is necessary to verify that the ARIMA residuals $\hat{\varepsilon}_t$ actually exhibit conditional heteroskedasticity, that is, that their variance is not constant over time. This is established using the ARCH Lagrange Multiplier (ARCH-LM) test proposed by Engle [4].

The test regresses the squared residuals on their own lags:

$$\hat{\varepsilon}_t^2 = a_0 + \sum_{i=1}^{m} a_i \hat{\varepsilon}_{t-i}^2 + u_t, \tag{4}$$

and tests $H_0 : a_1 = a_2 = \cdots = a_m = 0$ (no ARCH effect, i.e. constant variance). The LM statistic $T \cdot R^2$ follows a $\chi^2(m)$ distribution asymptotically under H_0, where m is the number of lags. In this study, $m = 10$ lags are used. Rejecting H_0 confirms that the squared residuals are serially correlated, which means that the variance is not constant and that a GARCH-type model is statistically warranted.

Figures 5–7 provide a visual three-panel diagnostic of the ARIMA(4, 0, 0) residuals that complements the formal ARCH-LM test.

Figure 5 plots the ARIMA residuals over the training period. The series fluctuates around zero with no discernible trend, confirming that the mean equation has been adequately specified. However, the dispersion of the residuals is clearly non-constant: stretches of small residuals in low-volatility regimes alternate with episodes of large positive and negative spikes, most prominently during the COVID-19 crash of March 2020. This visual pattern of *volatility clustering* in the residuals is the direct motivation for applying a GARCH variance model.

Figure 6 plots the squared residuals $\hat{\varepsilon}_t^2$, which serve as a proxy for daily variance. The dominance of a single spike near 2020 (squared residual exceeding 125) makes the heteroskedasticity in the variance process unmistakable: the squared residuals are far from the constant level that would be expected under homoskedasticity, and their behavior is clearly driven by time-varying volatility regimes.

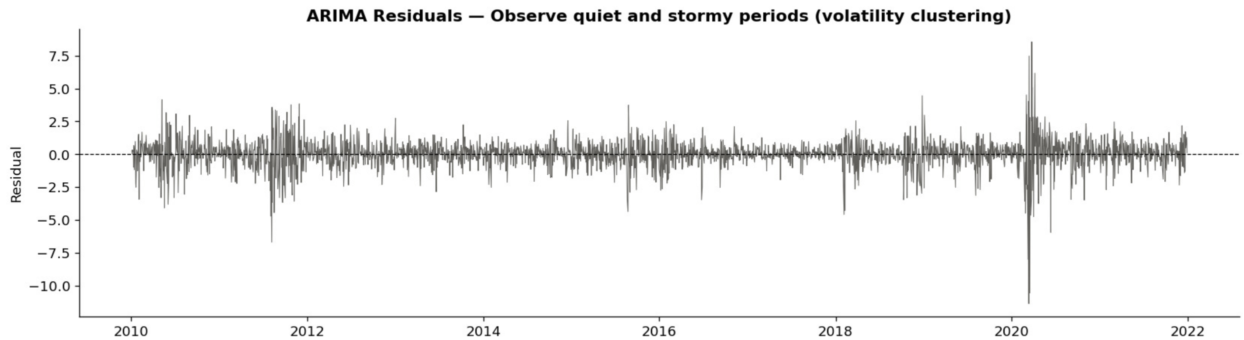

Fig. 5 ARIMA(4, 0, 0) residuals over the training set (January 2010 – December 2021). The residuals fluctuate around zero (confirming adequate mean specification) but exhibit clear heteroskedasticity: tranquil periods of small residuals alternate with volatile episodes, particularly the COVID-19-induced spike in early 2020 (residual exceeding -10%). This volatility clustering pattern motivates GARCH modelling of the conditional variance.

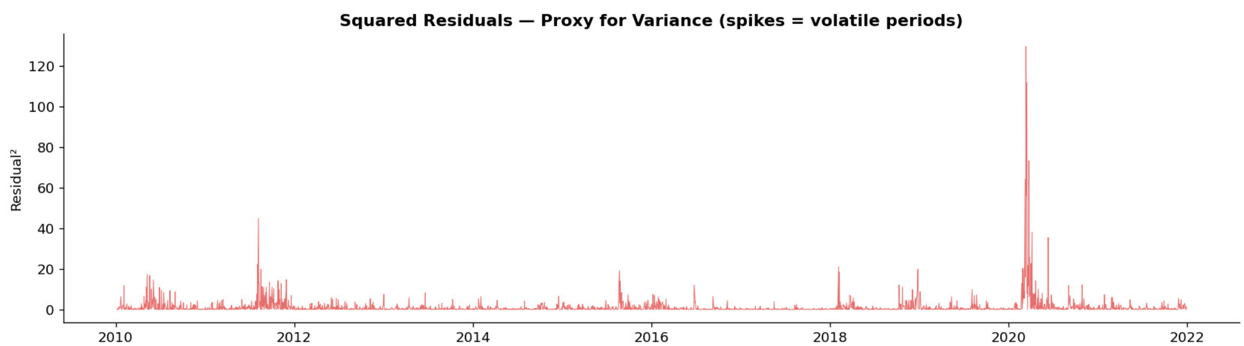

Fig. 6 Squared ARIMA(4, 0, 0) residuals over the training set (January 2010 – December 2021), serving as a proxy for daily variance. The extreme spike in early 2020 (reaching approximately 128) corresponds to the COVID-19 market dislocation and dwarfs all other variance episodes in the sample. The pronounced non-constancy of the squared residuals confirms the presence of conditional heteroskedasticity.

Figure 7 plots the ACF of the squared residuals up to 30 lags. In sharp contrast to the ACF of the level returns (Figure 4), virtually every bar in this plot extends far beyond the 95% confidence band (red dashed line), with autocorrelations at short lags exceeding 0.50. This strong and slowly decaying serial dependence in the squared residuals is the defining signature of an ARCH process and constitutes graphical confirmation of the ARCH-LM test result. The slow decay also suggests that volatility shocks are highly persistent, which motivates the use of the GARCH(1,1) specification rather than a finite-order ARCH model.

D. GARCH-Family Models

Six GARCH-family models are estimated, formed by the full factorial combination of three variance specifications and two conditional error distributions, as summarized in Table 2.

1. GARCH(1,1)

The baseline model is the symmetric GARCH(1,1) of Bollerslev [5], whose variance equation is:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{5}$$

where $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and the covariance-stationarity constraint $\alpha + \beta < 1$ must hold. The ARCH coefficient α measures the sensitivity of current variance to the most recent squared shock: a larger α implies that a sudden spike in returns today has a stronger and more immediate effect on tomorrow's perceived risk. The GARCH coefficient

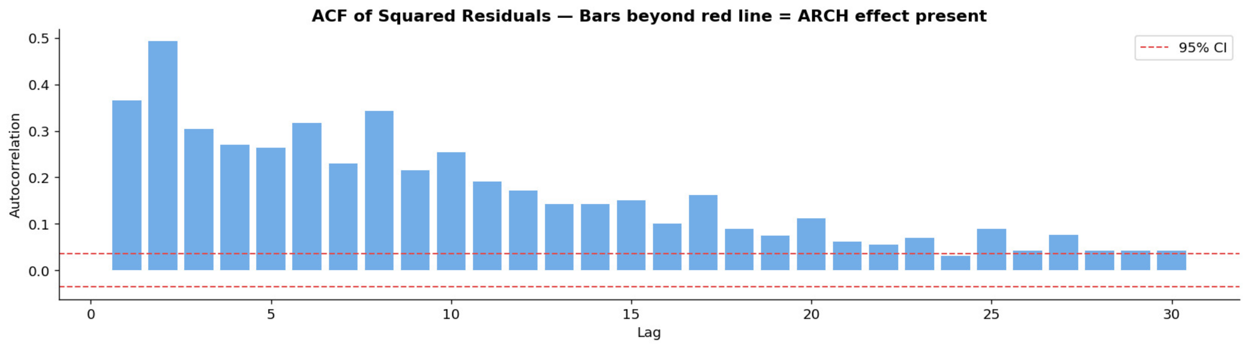

Fig. 7 ACF of squared ARIMA(4, 0, 0) residuals (training set, up to 30 lags). The red dashed lines mark the 95% confidence interval under the null of zero autocorrelation. All bars at short lags substantially exceed the confidence band (autocorrelation at lag 2 exceeds 0.50), confirming strong and persistent serial dependence in the squared residuals. This pattern is consistent with an ARCH process and statistically justifies the use of GARCH-family models for the conditional variance. ARCH-LM(χ^2) = 982.20, $p < 0.001$.

Table 2 Summary of All Models Estimated. The seven candidate models comprise six GARCH-family specifications and the HAR-RV benchmark. For GARCH models, $(\omega, \alpha, \beta, \gamma, \nu)$ denote the intercept, ARCH coefficient, GARCH coefficient, asymmetry parameter, and Student-t degrees of freedom, respectively. For HAR-RV, $(c, \beta_d, \beta_w, \beta_m)$ denote the intercept and the daily, weekly, and monthly coefficients.

Model	Distribution	Parameters	Key Feature
GARCH(1,1)	Normal	ω, α, β	Symmetric variance
GARCH(1,1)	Student-t	$\omega, \alpha, \beta, \nu$	Fat tails
GJR-GARCH(1,1)	Normal	$\omega, \alpha, \beta, \gamma$	Leverage effect
GJR-GARCH(1,1)	Student-t	$\omega, \alpha, \beta, \gamma, \nu$	Leverage + fat tails
EGARCH(1,1)	Normal	$\omega, \alpha, \beta, \gamma$	Log-variance, asymmetric
EGARCH(1,1)	Student-t	$\omega, \alpha, \beta, \gamma, \nu$	Log-variance + fat tails
HAR-RV (OLS)	–	$c, \beta_d, \beta_w, \beta_m$	Multi-horizon RV

β measures persistence: a value close to one implies that elevated variance decays very slowly, consistent with the long-lived volatility swings observed in equity markets. The sum $(\alpha + \beta)$ is the total persistence parameter; values close to, but below, unity imply that shocks die out gradually rather than permanently.

2. GJR-GARCH(1,1)

The GJR-GARCH model of Glosten, Jagannathan, and Runkle [6] augments the GARCH variance equation with an indicator function to capture the leverage effect:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\,\varepsilon_{t-1}^2\,\mathbf{1}[\varepsilon_{t-1} < 0] + \beta\sigma_{t-1}^2, \tag{6}$$

where $\mathbf{1}[\varepsilon_{t-1} < 0]$ equals one when the lagged innovation is negative (bad news) and zero otherwise. The parameter γ directly quantifies the leverage effect: a positive and statistically significant γ confirms that negative return shocks amplify variance by $(\alpha + \gamma)$, compared with only α for a positive shock of equal size. This asymmetric response reflects the empirical observation that falling equity prices raise a firm's financial leverage and heighten investor uncertainty more than gains of the same magnitude.

3. EGARCH(1,1)

Nelson's [7] Exponential GARCH model parameterizes the logarithm of conditional variance:

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}, \tag{7}$$

where the term $|\varepsilon_{t-1}|/\sigma_{t-1}$ captures the magnitude of the standardized shock and the sign term $\gamma\, \varepsilon_{t-1}/\sigma_{t-1}$ captures the direction. A negative value of γ implies that negative standardized shocks raise log-variance by more than positive shocks of equal magnitude, which is the EGARCH representation of the leverage effect. The log-variance formulation has two practical advantages over the level-variance GARCH: it guarantees $\sigma_t^2 > 0$ for any parameter values without imposing explicit non-negativity constraints, and it can represent more complex asymmetric shapes in the news impact curve.

4. Conditional Error Distributions

Each of the three variance specifications above is estimated under two conditional error distributions, yielding six models in total.

Under the **Normal** (Gaussian) distribution, the standardized innovations $z_t = \varepsilon_t/\sigma_t$ are assumed i.i.d. $\mathcal{N}(0, 1)$.

Under the **Student-*t*** distribution [8], $z_t \sim t(\nu)$, where the degrees-of-freedom parameter $\nu > 2$ is estimated from the data jointly with the variance equation parameters. A smaller ν implies heavier tails: the distribution assigns higher probability to extreme return realizations relative to the normal. Values of $\nu < 10$ are generally taken as evidence of economically meaningful excess tail risk that the Gaussian assumption would understate. As documented in Section III, the excess kurtosis of 13.21 in the return data provides strong prior evidence that ν will be estimated well below 10, and that the Student-*t* models should produce materially better-calibrated VaR estimates in the tails.

All six GARCH models are estimated by Maximum Likelihood on the training set using the `arch` Python library. Model selection is based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), both of which penalize log-likelihood for the number of estimated parameters:

$$\text{AIC} = -2 \ln \hat{L} + 2k, \qquad \text{BIC} = -2 \ln \hat{L} + k \ln T, \tag{8}$$

where $\hat{L}$ is the maximized likelihood, k is the number of free parameters, and T is the sample size. Lower values of AIC and BIC indicate a better balance between fit and parsimony.

E. Realized Volatility Proxies

A key methodological contribution of this study is the systematic evaluation of four realized volatility proxies within the HAR-RV framework. All proxies are constructed from daily OHLC price data retrieved from Yahoo Finance and serve as both the dependent variable in the HAR-RV regression and the inputs to the multi-horizon lag components.

Absolute Return (baseline), following Ding et al. [12]:

$$\widehat{\sigma}_t^{\text{abs}} = |r_t|. \tag{9}$$

This estimator requires only daily closing prices but discards all intraday variation and is the noisiest of the four proxies considered.

Parkinson Estimator [13]:

$$\widehat{\sigma}_t^{\text{Park}} = \frac{[\ln(H_t/L_t)]^2}{4 \ln 2}, \tag{10}$$

where H_t and L_t denote the daily high and low prices. Under geometric Brownian motion without drift and no overnight gaps, this estimator is approximately four to five times more efficient than the close-to-close estimator, as it utilises the full intraday price range.

Garman-Klass Estimator [14]:

$$\widehat{\sigma}_t^{\text{GK}} = \frac{1}{2} \left[\ln\!\left(\frac{H_t}{L_t} \right) \right]^2 - (2 \ln 2 - 1) \left[\ln\!\left(\frac{C_t}{O_t} \right) \right]^2, \tag{11}$$

where O_t and C_t denote the daily open and close. By incorporating the opening price alongside the intraday range, this estimator achieves approximately seven times the efficiency of the close-to-close estimator.

Yang-Zhang Estimator [15]:

$$\widehat{\sigma}_t^{\mathrm{YZ}} = \sigma^2_{\mathrm{overnight}} + k\,\sigma^2_{\mathrm{open\text{-}to\text{-}close}} + (1-k)\,\sigma^2_{\mathrm{RS}}, \tag{12}$$

where $k = 0.34 \,/\, \left[1.34 + (T+1)/(T-1)\right]$ and T is the number of observations in the estimation window. This estimator is drift-independent and combines overnight, intraday, and Rogers-Satchell components, making it theoretically optimal when returns exhibit both intraday drift and overnight jumps. The full derivation is provided in Yang and Zhang [15].

For evaluation purposes, each proxy-fitted HAR-RV model is assessed against a common target, the absolute return $|r_t|$, so that cross-proxy comparisons reflect model quality rather than proxy quality. Cross-proxy QLIKE evaluation, where each model is assessed against all four proxies as the realised volatility benchmark, is additionally reported in Appendix Tables 21— 22 to verify the robustness of the model ranking to the choice of evaluation target.

F. HAR-RV Model

The Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) of Corsi [10] serves as the semi-parametric benchmark. Since intraday transaction data are not available for this study, realized volatility is proxied by the absolute daily log return $\mathrm{RV}_t = |r_t|$, following the approach of Ding, Granger, and Engle [12].

The multi-horizon components are constructed as:

$$\mathrm{RV}_t^{(d)} = |r_t|, \tag{13}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{i=1}^{5} |r_{t-i+1}|, \tag{14}$$

$$\mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{i=1}^{22} |r_{t-i+1}|, \tag{15}$$

representing the daily, weekly (five-day), and monthly (twenty-two-day) realized volatility averages, respectively.

The HAR-RV regression is then:

$$\mathrm{RV}_{t+1} = c + \beta_d\,\mathrm{RV}_t^{(d)} + \beta_w\,\mathrm{RV}_t^{(w)} + \beta_m\,\mathrm{RV}_t^{(m)} + \varepsilon_{t+1}, \tag{16}$$

estimated by Ordinary Least Squares (OLS). The economic interpretation of the three slope coefficients is direct. The daily coefficient β_d reflects the persistence of short-run shocks generated by high-frequency traders. The weekly coefficient β_w captures medium-horizon momentum from speculative activity. The monthly coefficient β_m embeds the slow-moving risk perceptions of institutional and long-term investors. Despite the simplicity of OLS estimation, the model generates long-memory-like autocorrelation in its volatility forecasts through this cascade of heterogeneous components, which is the key mechanism behind its competitive forecasting performance [10].

G. Expanding-Window Out-of-Sample Forecasting and Loss Functions

All out-of-sample volatility forecasts are produced using an *expanding-window* scheme. At each step t in the test period, the model is re-estimated on all available data from the beginning of the sample up to and including day t, and a one-step-ahead forecast $\hat{\sigma}_{t+1}$ (or $\widehat{\mathrm{RV}}_{t+1}$ for HAR-RV) is produced for day $t+1$. The estimation window then expands by one observation, and the process repeats for every test day. This procedure is illustrated schematically as follows:

$$\begin{aligned}
&\text{Step 1:}\quad \text{Fit on } [1, T_0] \to \text{Forecast } \hat{\sigma}_{T_0+1}\\
&\text{Step 2:}\quad \text{Fit on } [1, T_0+1] \to \text{Forecast } \hat{\sigma}_{T_0+2}\\
&\quad\vdots\\
&\text{Step } n:\quad \text{Fit on } [1, T_0+n-1] \to \text{Forecast } \hat{\sigma}_{T_0+n}
\end{aligned}$$

where $T_0 = 3{,}017$ is the initial training size.

The expanding-window scheme is chosen over the alternative rolling-window approach for two reasons. First, it exploits the maximum amount of historical information at every forecast origin, which is generally more efficient when the data-generating process is stable. Second, it mirrors the approach a practitioner would follow in a real-time risk management setting, where all historical data are retained and used to continuously update model estimates. Crucially, the *identical* methodology is applied to both the GARCH family and the HAR-RV benchmark, ensuring that the comparison is not contaminated by differences in the estimation scheme.

Forecast accuracy in the out-of-sample period is evaluated using three standard loss functions. Let $\hat{\sigma}_t$ denote the volatility forecast and $\mathrm{RV}_t = |r_t|$ denote the realized volatility proxy:

$$\mathrm{MSE} = \frac{1}{T}\sum_{t=1}^{T}(\hat{\sigma}_t - \mathrm{RV}_t)^2, \tag{17}$$

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T}|\hat{\sigma}_t - \mathrm{RV}_t|, \tag{18}$$

$$\mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\ln \hat{\sigma}_t^2 + \frac{\mathrm{RV}_t^2}{\hat{\sigma}_t^2}\right). \tag{19}$$

MSE penalizes large forecast errors disproportionately and is therefore sensitive to outliers during stress episodes. MAE treats all errors equally and provides a more robust measure of average forecast deviation. QLIKE is an asymmetric loss function derived from the Gaussian quasi-likelihood: it penalizes under-prediction of volatility more heavily than over-prediction, making it particularly relevant for risk management applications where the cost of underestimating risk exceeds the cost of overestimating it.

Statistical significance of forecast differences. To assess whether observed out-of-sample loss differences reflect genuine model superiority or sampling variation, all pairwise forecast comparisons are evaluated using the Diebold–Mariano (DM) test [20]. For benchmark model i and comparator model j, the loss differential is $d_t = L(\hat{\sigma}_{i,t}, \mathrm{RV}_t) - L(\hat{\sigma}_{j,t}, \mathrm{RV}_t)$, and the DM statistic

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\sigma}_d^2/T}} \xrightarrow{d} \mathcal{N}(0,1) \tag{20}$$

tests $H_0 : \mathbb{E}[d_t] = 0$ (equal predictive accuracy) against a two-sided alternative. The long-run variance $\widehat{\sigma}_d^2$ is estimated by the Newey–West HAC estimator with $h = 1$ lag. Since six pairwise tests are run for each loss function, the Holm–Bonferroni sequential correction [21] is applied to control the family-wise error rate at $\alpha = 0.05$. HAR-RV serves as the benchmark (Model 1) in all comparisons; a positive DM statistic indicates the GARCH comparator achieves lower loss.

H. Value-at-Risk and Expected Shortfall

1. Parametric VaR and ES Formulas

Under the parametric approach, the one-day-ahead VaR at confidence level $1 - \alpha$ is computed directly from the conditional volatility forecast:

$$\mathrm{VaR}_{t+1}(\alpha) = \hat{\mu}_{t+1} + F_z^{-1}(\alpha)\cdot\hat{\sigma}_{t+1}, \tag{21}$$

where $\hat{\mu}_{t+1}$ is the one-step-ahead conditional mean from the ARIMA equation (set to zero for HAR-RV), $\hat{\sigma}_{t+1}$ is the conditional volatility forecast, and $F_z^{-1}(\alpha)$ is the α-quantile of the assumed standardized residual distribution. For the Normal distribution, $F_z^{-1}(\alpha) = \Phi^{-1}(\alpha)$; for the Student-$t$ with $\hat{\nu}$ degrees of freedom, $F_z^{-1}(\alpha) = t_{\hat{\nu}}^{-1}(\alpha)$. Since $\alpha < 0.5$, both quantiles are negative, and VaR is reported as a negative number representing the lower bound of the return distribution.

The Expected Shortfall at confidence level $1 - \alpha$ is computed as:

$$\mathrm{ES}_{t+1}(\alpha) = \hat{\mu}_{t+1} - \frac{f_z\big(F_z^{-1}(\alpha)\big)}{\alpha}\cdot\hat{\sigma}_{t+1}, \tag{22}$$

where $f_z(\cdot)$ is the probability density function of the standardized residual distribution. For the Student-t, the ES expression becomes:

$$\mathrm{ES}_{t+1}(\alpha) = \hat{\mu}_{t+1} - \frac{f_t\left(t_{\hat{\nu}}^{-1}(\alpha);\ \hat{\nu}\right) \cdot \left(\hat{\nu} + \left[t_{\hat{\nu}}^{-1}(\alpha)\right]^2\right)}{\alpha\,(\hat{\nu} - 1)} \cdot \hat{\sigma}_{t+1}, \tag{23}$$

where $f_t(\cdot\,;\ \hat{\nu})$ is the Student-t density. ES is always more negative than VaR, reflecting the fact that it measures the average loss conditional on the loss exceeding the VaR threshold, and therefore always provides a more conservative risk estimate. VaR and ES are computed at both the 95% and 99% confidence levels.

2. HAR-RV-Based VaR

For the HAR-RV model, which does not directly produce a parametric distribution, the volatility forecast $\widehat{\mathrm{RV}}_{t+1}$ is used as the conditional standard deviation $\hat{\sigma}_{t+1}$ in equations (21)–(22), with the Normal distribution assumed for the standardized residuals. This approach is consistent with the semi-parametric interpretation of the HAR-RV and allows its VaR estimates to be evaluated on a directly comparable basis with those of the GARCH-Normal models.

I. Backtesting Procedures

1. Kupiec Proportion of Failures Test

The Kupiec [2] Proportion of Failures (POF) test assesses whether the empirical VaR violation rate is statistically consistent with the nominal confidence level. A violation on day t is defined as:

$$I_t = \mathbf{1}[r_t < \mathrm{VaR}_t(\alpha)]\,, \tag{24}$$

where $I_t = 1$ if the realized return falls below the VaR threshold (a violation) and $I_t = 0$ otherwise. Let $N = \sum_t I_t$ be the total number of violations over T test observations, and $\hat{p} = N/T$ the empirical violation rate. The null hypothesis is $H_0 : \hat{p} = \alpha$ (the observed violation rate equals the nominal level). The likelihood ratio test statistic is:

$$\mathrm{LR}_{\mathrm{uc}} = -2\ln\frac{(1-\alpha)^{T-N}\,\alpha^N}{(1-\hat{p})^{T-N}\,\hat{p}^N} \xrightarrow{d} \chi^2(1), \tag{25}$$

under H_0. A model with too many violations ($\hat{p} > \alpha$) underestimates risk; one with too few ($\hat{p} < \alpha$) is over-conservative. Both cases fail the test when p-value < 0.05.

2. Christoffersen Conditional Coverage Test

The Christoffersen [3] Conditional Coverage (CC) test jointly evaluates unconditional coverage and the independence of violations. A model may produce the correct average violation rate but still be misspecified if violations cluster in time (for instance, occurring on consecutive trading days during a stress episode), which would indicate that the model fails to adapt quickly enough to changing volatility regimes.

The test models the violation indicator sequence $\{I_t\}$ as a first-order Markov chain, estimating the transition probabilities:

$$\hat{\pi}_{ij} = \frac{n_{ij}}{\sum_j n_{ij}}, \quad i, j \in \{0, 1\}, \tag{26}$$

where n_{ij} is the number of transitions from state i to state j. Under the null hypothesis of independence, $\pi_{01} = \pi_{11}$, meaning the probability of a violation is the same regardless of whether a violation occurred on the previous day. The likelihood ratio statistic for independence is:

$$\mathrm{LR}_{\mathrm{ind}} = -2\ln\frac{(1-\hat{\pi})^{n_{00}+n_{10}}\hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\hat{\pi}_{01}^{n_{01}}(1-\hat{\pi}_{11})^{n_{10}}\hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi^2(1), \tag{27}$$

where $\hat{\pi} = (n_{01} + n_{11})/(n_{00} + n_{01} + n_{10} + n_{11})$ is the unconditional violation rate. The joint Conditional Coverage statistic is:

$$\mathrm{LR}_{\mathrm{cc}} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{ind}} \xrightarrow{d} \chi^2(2). \tag{28}$$

A well-specified VaR model must pass both the Kupiec and Christoffersen tests at both the 95% and 99% confidence levels. Passing the Kupiec test alone is insufficient: a model that generates the correct average number of violations but clusters them during periods of acute market stress – such as the COVID-19 crash or the 2022 bear market – provides a misleading picture of risk and fails to satisfy the independence requirement essential for regulatory compliance under Basel III [17].

V. Empirical Results

This section presents and interprets the empirical findings across all dimensions of the analysis, proceeding in the order of the modelling pipeline established in Section IV. The discussion covers stationarity and ARIMA diagnostics, GARCH in-sample estimation and model selection, HAR-RV estimation, out-of-sample forecast accuracy, VaR and ES calibration, and formal backtesting.

A. Stationarity Tests and ARIMA Results

1. Stationarity Tests

Table 3 reports the ADF and KPSS test results for the raw closing-price series and the log-return series.

Table 3 Stationarity Test Results for S&P 500 Closing Price and Log Returns. ADF null hypothesis: unit root (non-stationary); reject H_0 implies stationary. KPSS null hypothesis: stationary; failure to reject H_0 implies stationary.

Series	ADF Stat	ADF p-value	KPSS Stat	Conclusion
Closing price	1.4323	0.9973	9.0598	Non-stationary
Log return	−13.3438	< 0.001	0.0274	Stationary

The raw closing-price series is clearly non-stationary: the ADF statistic of 1.43 fails to reject the unit-root null ($p = 0.997$), and the KPSS statistic of 9.06 strongly rejects the stationarity null. This is the expected outcome for an equity price index, which follows an upward trend over time. The log-return series, by contrast, is unambiguously stationary: the ADF statistic of −13.34 ($p < 0.001$) decisively rejects the unit-root null, and the KPSS statistic of 0.027 fails to reject stationarity ($p \geq 0.10$). Both tests thus agree, providing the strongest possible joint evidence that log returns are $I(0)$ stationary, as required for valid ARIMA and GARCH estimation.

2. ARIMA Mean Equation

The `auto_arima` stepwise search over AIC selected **ARIMA(4, 0, 0)** as the optimal mean specification (AIC = 8,972.27; BIC = 9,008.34), representing an improvement of 12.35 points over the next-best specification, ARIMA(1, 0, 1), and 90.58 points over a constant mean equation. All four AR coefficients are statistically significant at the 5% level, with estimates $\hat{\phi}_1 = -0.137$ ($p < 0.001$), $\hat{\phi}_2 = +0.083$ ($p < 0.001$), $\hat{\phi}_3 = -0.018$ ($p = 0.040$), and $\hat{\phi}_4 = -0.061$ ($p < 0.001$). The alternating signs indicate a mild short-run mean-reversion pattern, consistent with bid–ask bounce and short-horizon return reversal documented in the equity microstructure literature. Although the magnitudes are modest, extracting this predictable mean structure ensures that GARCH models are applied to the pure innovation sequence rather than to autocorrelation-contaminated returns. The selection of ARIMA(4,0,0) is confirmed by a stepwise AIC comparison over nested specifications reported in Appendix Table 18; all pass/fail conclusions are robust to a constant mean equation specification, confirming that the choice of mean equation does not drive the main findings.

3. ARCH Effect Test

The ARCH-LM test results are reported in Table 4. The χ^2 statistic of 982.20 ($p < 0.001$) overwhelmingly rejects the null hypothesis of no ARCH effect. This finding confirms that the variance of the ARIMA residuals is far from constant over time and that a GARCH-type model is statistically warranted. The visual evidence presented in Section IV (Figures 5–7) corroborates this conclusion: squared residuals exhibit strongly persistent autocorrelation at all lags up to 30, with autocorrelations at short lags exceeding 0.50.

Table 4 ARCH-LM Test Results for ARIMA(4, 0, 0) Residuals (10 lags). H_0: no ARCH effect (constant variance). Rejection confirms conditional heteroskedasticity and justifies GARCH modelling.

Test	Statistic	p-value	Conclusion
ARCH-LM (χ^2)	982.20	< 0.001	ARCH effect confirmed
F-statistic	145.33	< 0.001	ARCH effect confirmed

B. GARCH Model Estimation

1. In-Sample Fit and Model Selection

Table 5 reports the estimated parameters and information criteria for all six GARCH-family models, estimated by Maximum Likelihood on the training set (T = 3,017 observations).

Table 5 GARCH-Family Model Estimation Results (Training Set, January 2010 – December 2021). Parameters $(\omega, \alpha, \beta, \gamma, \nu)$ are the variance intercept, ARCH coefficient, GARCH coefficient, asymmetry parameter, and Student-t degrees of freedom, respectively. Persistence = $\alpha + \beta$ for GARCH/GJR; reported as $|\beta|$ from the log-variance equation for EGARCH. The best model by AIC and BIC is EGARCH(1,1)-Student-t.

Model	Dist.	ω	α	β	γ	ν	AIC	BIC	Persist.
GARCH(1,1)	Normal	0.0421***	0.1887***	0.7745***	–	–	7531.25	7555.30	0.963
GARCH(1,1)	Student-t	0.0293***	0.1924***	0.7973***	–	4.999***	7320.70	7350.76	0.990
GJR-GARCH(1,1)	Normal	0.0429***	0.0439	0.7854***	0.2590***	–	7432.96	7463.02	0.829
GJR-GARCH(1,1)	Student-t	0.0330***	0.0000	0.8066***	0.3353***	5.323***	7202.80	7238.87	0.807
EGARCH(1,1)	Normal	−0.0091	0.2484***	0.9426***	−0.1743***	–	7408.99	7439.05	1.191
EGARCH(1,1)	**Student-t**	−0.0118†	0.2047***	0.9594***	−0.2107***	5.491***	**7191.03**	**7227.10**	1.164

*** significant at 1%; † p = 0.063 (marginally insignificant at 5%).

Persistence = $\alpha + \beta$ for GARCH and GJR-GARCH; Persistence = $|\beta|$ for EGARCH (values above 1 are permissible).

GJR-GARCH(1,1)-Student-t: α estimate is effectively zero (numerical boundary); leverage effect fully captured by $\gamma = 0.3353^{***}$.

Standardised residual diagnostics confirm ARCH-LM p > 0.54 for all specifications, indicating adequate variance specification (full diagnostics in Appendix Table 17)

Figure 8 displays the AIC and BIC values for all six models as horizontal bar charts, making the ranking immediately visible. Both criteria agree unambiguously: the three Student-t models occupy the bottom of the ranking (lowest AIC/BIC), and EGARCH(1,1)-Student-t is the clear winner.

The results reveal a clear pattern: Student-t specifications uniformly dominate their Gaussian counterparts within each model class. Switching from Normal to Student-t reduces the AIC by 210.5 points for GARCH(1,1), by 230.2 points for GJR-GARCH(1,1), and by 218.0 points for EGARCH(1,1), confirming the strong empirical relevance of fat-tailed return innovations in S&P 500 daily returns. Within the Student-t family, asymmetric specifications further outperform the symmetric baseline, with AIC improvements of 117.9 and 129.7 points for GJR-GARCH(1,1) and EGARCH(1,1) respectively, providing formal statistical support for the leverage effect hypothesis.

The boundary solution $\hat{\alpha} = 0.000$ obtained for GJR-GARCH(1,1)-Student-t warrants further examination. A restricted specification imposing $\alpha \equiv 0$ yields a log-likelihood of −3,621.66 against −3,621.62 for the unrestricted model, producing a Likelihood Ratio statistic of 0.078 ($p = 0.780$) fails to reject the null that $\alpha = 0$, and the restricted model achieves lower information criteria, indicating that the leverage coefficient $\hat{\gamma} = 0.303$ adequately captures the asymmetric response without requiring an independent ARCH term. The LR statistic of 0.078 ($p = 0.780$) fails to reject the null that $\alpha = 0$, and the restricted model achieves lower information criteria, indicating that the leverage coefficient $\hat{\gamma} = 0.303$ adequately captures the asymmetric response without requiring an independent ARCH term. Results from Table 24 are consistent with this interpretation.

The best model by both AIC (7,191.03) and BIC (7,227.10) is the **EGARCH(1,1)-Student-t**. Its estimated parameters carry the following economic interpretations. The ARCH coefficient $\hat{\alpha} = 0.205$ ($p < 0.001$) indicates that the standardized magnitude of yesterday's shock has a substantial immediate effect on today's variance. The GARCH coefficient $\hat{\beta} = 0.959$ ($p < 0.001$) reflects very high persistence: a volatility shock decays by only 4.1% per day, implying that elevated volatility takes approximately three to four months to revert to its long-run mean. The asymmetry parameter $\hat{\gamma} = -0.211$ ($p < 0.001$) is negative and highly significant, confirming the leverage effect: negative return

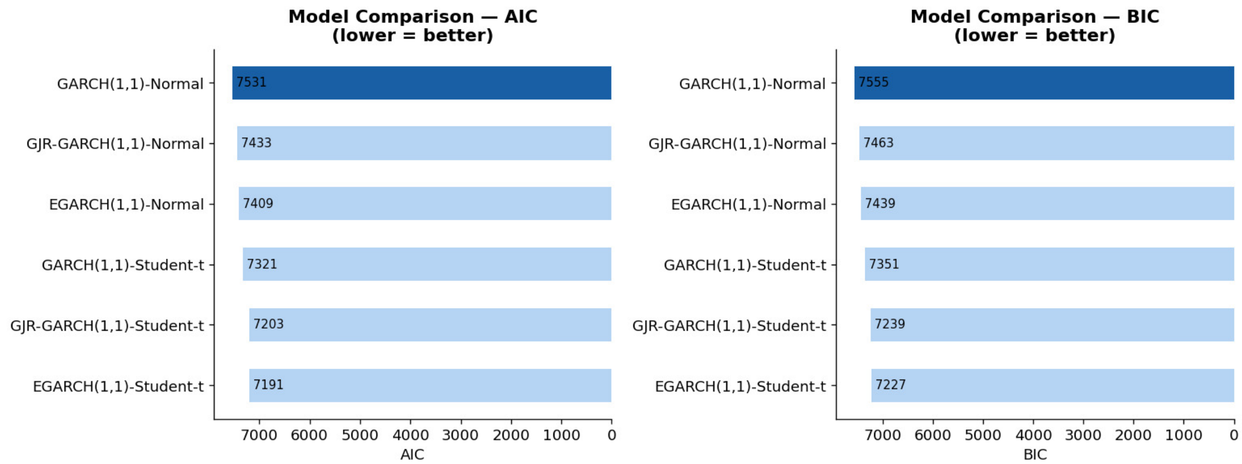

Fig. 8 GARCH-Family Model Comparison by AIC and BIC (Training Set, January 2010 – December 2021). ***Left panel:*** **AIC values (lower = better in-sample fit penalised for parameters).** ***Right panel:*** **BIC values (stricter penalty for additional parameters). Both criteria rank EGARCH(1,1)-Student-t first (AIC = 7,191; BIC = 7,227) and GARCH(1,1)-Normal last (AIC = 7,531; BIC = 7,555). The consistent ranking across both criteria confirms that the fat-tailed and asymmetric specifications are not merely over-parameterised fits.**

shocks increase log-variance by (-0.211) additional units compared with positive shocks of equal magnitude, consistent with the well-documented finding that bad news amplifies equity market volatility more than good news. The estimated degrees-of-freedom parameter $\hat{\nu} = 5.49$ ($p < 0.001$) is well below 10, confirming economically meaningful fat tails that could not be captured by the Gaussian assumption. This value implies that the conditional return distribution assigns substantially more probability to extreme losses than a normal distribution, a property that is directly relevant to the accuracy of VaR estimates at high confidence levels.

Figure 9 displays the conditional volatility estimates from the best model (EGARCH(1,1)-Student-t) over the full sample, with key market stress episodes highlighted. The model captures the sharp COVID-19 spike to $\approx 2.6\%$ in March 2020 and the elevated but more gradual volatility regime of 2022, demonstrating its ability to respond to both acute and persistent stress episodes. The leverage effect parameter $\hat{\gamma} = -0.211$ enables the model to react more sharply to negative shocks than to positive ones of equal magnitude, which is clearly visible in the asymmetric spikes around the two stress periods.

C. HAR-RV Model Results

Table 6 reports the OLS estimates for the HAR-RV model estimated on the training set.

Table 6 HAR-RV OLS Regression Results (Training Set, January 2010 – December 2021, $N = 2{,}995$ observations after lag construction). Dependent variable: $\mathrm{RV}_{t+1} = |r_{t+1}|$. Standard errors are heteroskedasticity-robust.

Component	Coefficient	Std. Error	t-stat	p-value
Intercept (c)	0.1377	0.0225	6.11	< 0.001
β_d (Daily)	−0.0105	0.0221	−0.48	0.635
β_w (Weekly)	0.6201	0.0415	14.96	< 0.001
β_m (Monthly)	0.1922	0.0415	4.63	< 0.001
R^2	0.281			
F-statistic	390.2 ($p < 0.001$)			

Several findings are noteworthy. First, the overall model is highly significant ($F = 390.2$, $p < 0.001$), confirming that the multi-horizon structure of the HAR-RV captures genuine predictive information about future realized volatility. The $R^2 = 0.281$ indicates that 28.1% of next-day absolute return variation is explained by its own history at three

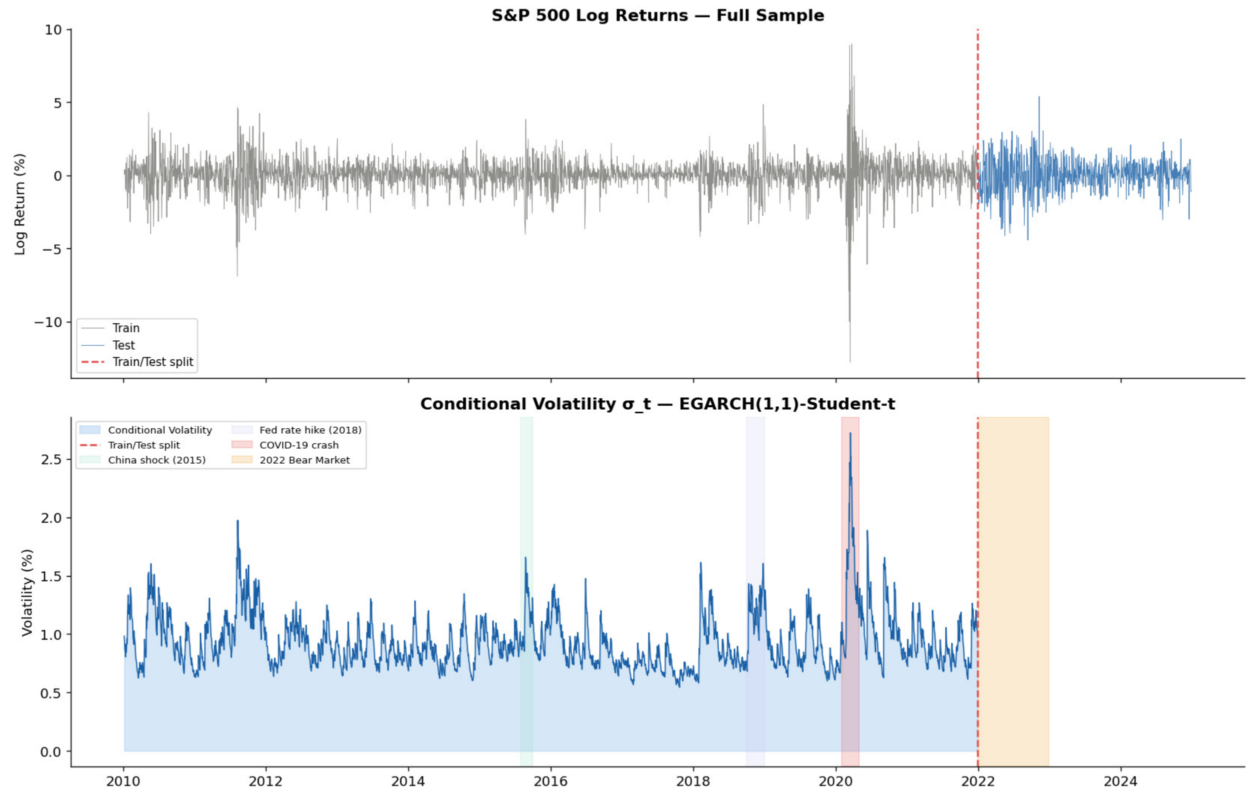

Fig. 9 S&P 500 Log Returns and Conditional Volatility $\hat{\sigma}_t$ from EGARCH(1,1)-Student-*t* (Full Sample, January 2010 – December 2024). *Upper panel:* Daily log returns; grey = training set, blue = test set; red dashed line marks the train/test split (December 2021). *Lower panel:* Estimated conditional volatility $\hat{\sigma}_t$ (%) with shading for the China shock (2015, green), Fed rate-hiking cycle (2018, blue), COVID-19 crash (2020, pink), and 2022 bear market (orange). The model clearly resolves all four stress regimes as distinct volatility elevations against an otherwise calm baseline.

horizons, which is a substantial proportion for a one-step-ahead forecast of a noisy daily series.

Second, the dominant forecasting contribution comes from the **weekly component**, with $\hat{\beta}_w = 0.621$ ($p < 0.001$). This coefficient implies that a one-unit increase in average weekly volatility is associated with a 0.621 increase in tomorrow's realized volatility, holding the other components constant. The large and significant weekly coefficient supports the Heterogeneous Market Hypothesis: the primary volatility signal comes from participants with a multi-day horizon (speculative traders and short-term fund managers), not from intraday noise.

Third, the **monthly component** is also significant ($\hat{\beta}_m = 0.192$, $p < 0.001$), capturing the slow-moving volatility expectations of institutional investors. Together, $\hat{\beta}_w + \hat{\beta}_m = 0.812$, implying strong medium-to-long-horizon persistence in the volatility process.

Fourth, the **daily component** $\hat{\beta}_d = -0.010$ is not statistically significant ($p = 0.635$). This finding suggests that single-day idiosyncratic noise in $|r_t|$ has no independent predictive value once the weekly and monthly averages are controlled for. The practical implication is that, when using absolute returns as a realized volatility proxy, the HAR-RV effectively reduces to a two-component (weekly and monthly) specification.

D. Out-of-Sample Volatility Forecast Accuracy and Proxy Comparison

Table 7 reports the out-of-sample forecast accuracy metrics for all seven models over the 755-day test period (December 2021 – December 2024), produced using the expanding-window scheme described in Section IV.G. Table 8 extends the analysis to the four HAR-RV proxy variants. Figure 10 provides a visual overview of the forecasted conditional volatility paths from all six GARCH models over the test period.

The results present a nuanced picture that can be organised around three findings.

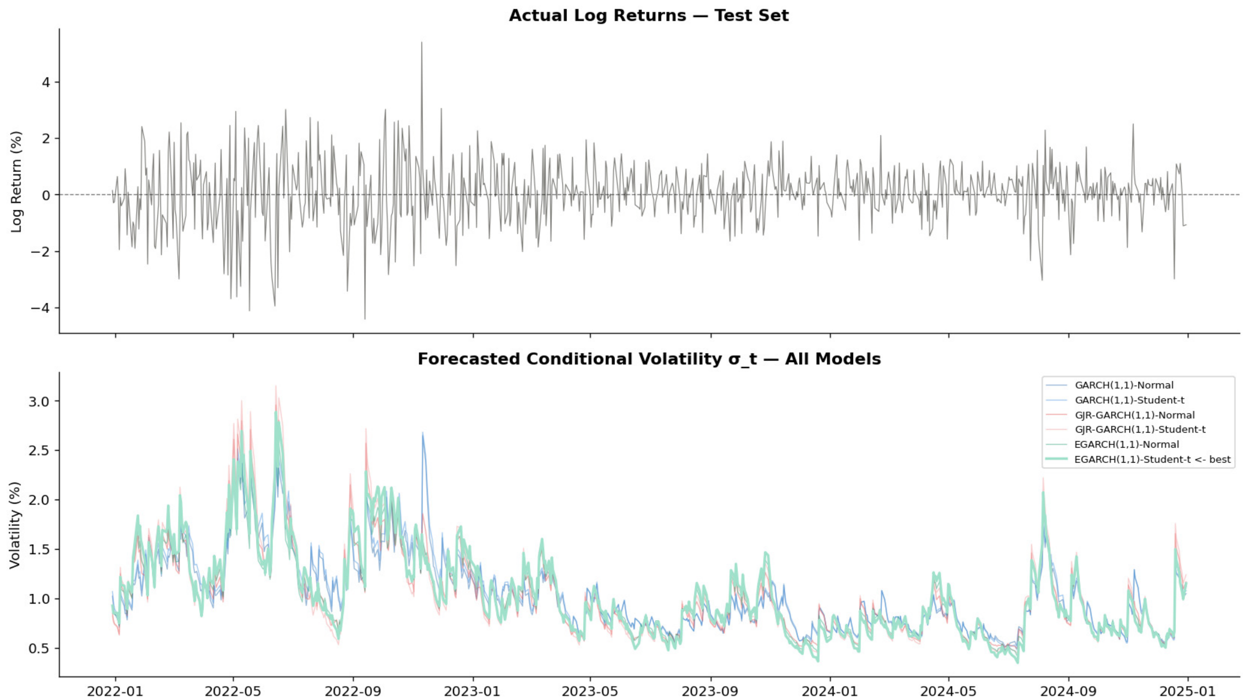

Fig. 10 Forecasted Conditional Volatility $\hat{\sigma}_t$ from All GARCH Models (Test Set, December 2021 – December 2024, $T = 755$ observations). ***Upper panel:*** **Actual daily log returns.** ***Lower panel:*** **One-step-ahead volatility forecasts from all six GARCH specifications. The EGARCH(1,1)-Student-t (best by AIC, highlighted in green) produces the most elevated forecasts during the 2022 bear market, while all models converge to similar low-volatility estimates during the calmer 2023–2024 recovery. The spread between models is largest during the stress period of early-to-mid 2022, where leverage-asymmetric models (GJR, EGARCH) diverge upward from the symmetric GARCH(1,1).**

1. Proxy-level findings: OHLC estimators improve HAR-RV point accuracy.

Under both MSE and MAE, OHLC-based proxies outperform the absolute-return baseline within the HAR-RV framework (Table 8). The Garman–Klass proxy achieves the lowest MSE (= 0.472), representing an 8.6% reduction relative to the absolute-return baseline (= 0.503), and ranks second under MAE (= 0.511). The Parkinson proxy achieves the lowest MAE (= 0.511) but ranks last under QLIKE (= 1.024). These improvements under symmetric loss are modest in absolute terms, but they confirm that the intraday range information embedded in OHLC data carries genuine predictive content not captured by close-to-close returns alone.

Under the QLIKE loss function, which penalises variance under-prediction asymmetrically, the proxy ranking reverses for Yang–Zhang: it achieves the lowest QLIKE (= 0.872) across all four proxy variants. Garman–Klass ranks second (= 0.925), while Parkinson ranks last (= 1.024). This divergence is consistent with the broader finding of this study: the Yang–Zhang estimator's overnight-gap adjustment provides the most relevant information precisely during elevated-risk periods, where variance under-prediction is most costly, even though this additional variability reduces its average-day accuracy under MSE and MAE.

Robustness analysis across five market regimes (Appendix Table 20) confirms that QLIKE and MAE rankings agree on the best proxy in only 25% of full-test-period regimes, rising to 100% agreement during extreme-volatility and low-volatility subperiods. This result reinforces the view that proxy selection recommendations should be conditioned on the intended loss criterion and application context.

2. Paradigm-level findings: HAR-RV wins under symmetric loss, GARCH wins under asymmetric loss.

Under both MSE and MAE, HAR-RV with the Garman–Klass proxy achieves the best out-of-sample performance among all specifications considered, with MAE = 0.511 compared with 0.586 for the best-performing GARCH model (EGARCH(1,1)-Normal), representing a 12.8% improvement. Even under the absolute-return proxy, HAR-RV achieves

Table 7 Out-of-Sample Forecast Accuracy: MSE, MAE, and QLIKE (Test Set, December 2021 – December 2024, $T = 755$ observations). Lower values indicate better forecast performance. Bold entries indicate the best value for each metric. Realized volatility proxy: $\mathrm{RV}_t = |r_t|$**.**

Model	Dist.	MSE	MAE	QLIKE
HAR-RV	–	**0.5041**	**0.5421**	1.3059
EGARCH(1,1)	Normal	0.5503	0.5857	**0.9881**
EGARCH(1,1)	Student-t	0.5679	0.5887	0.9912
GJR-GARCH(1,1)	Normal	0.5734	0.5910	1.0053
GJR-GARCH(1,1)	Student-t	0.5994	0.5965	1.0057
GARCH(1,1)	Normal	0.5743	0.6054	1.0283
GARCH(1,1)	Student-t	0.5936	0.6143	1.0244

Table 8 HAR-RV Volatility Proxy Comparison: Out-of-Sample Forecast Accuracy (Test Set, December 2021 – December 2024, $T = 755$ observations). Four realized-volatility proxies are evaluated as the dependent variable in the HAR-RV regression: the absolute daily log return $|r_t|$ (baseline), the Parkinson [13] range-based estimator, the Garman–Klass [14] open-high-low-close estimator, and the Yang–Zhang [15] overnight-adjusted estimator. All proxies are evaluated against a common target $|r_t|$ to allow a consistent apples-to-apples comparison across proxy types. Lower values indicate better forecast performance. Bold entries indicate the best value for each metric.

Proxy	MSE	MAE	QLIKE	MSE Rank	MAE Rank	QLIKE Rank
Garman–Klass (1980)	**0.4724**	0.5113	0.9251	1	2	2
Parkinson (1980)	0.4800	**0.5107**	1.0241	2	1	4
$|r_t|$ Abs. Return (baseline)	0.5028	0.5415	0.9572	3	3	3
Yang–Zhang (2000)	0.5238	0.5590	**0.8720**	4	4	1

Data requirements: Garman–Klass and Yang–Zhang require Open, High, Low, and Close prices; Parkinson requires High and Low only; $|r_t|$ requires closing prices only.

an 8.1% MAE improvement over EGARCH(1,1)-Normal. The pattern is consistent across all GARCH specifications: Student-t models are systematically outperformed by their Normal counterparts under both MSE and MAE, suggesting that the distributional assumption has limited impact on the point forecast of volatility but plays a more important role in tail risk quantification, as confirmed in the backtesting results in Section V.F.

Under the QLIKE loss function, the ranking reverses entirely. **EGARCH(1,1)-Normal achieves the lowest QLIKE** (= 0.988), followed closely by EGARCH(1,1)-Student-t (= 0.991), while HAR-RV with the absolute-return proxy produces the worst QLIKE (= 1.306) among all seven models. Importantly, this GARCH dominance under QLIKE is robust to proxy choice: cross-proxy evaluation in Appendix Tables 21–22 confirms that GARCH specifications occupy QLIKE ranks 1–6 across all four proxy evaluation targets, while all HAR-RV variants occupy ranks 7–10 regardless of which proxy serves as the benchmark. Among HAR-RV variants, even Yang–Zhang, the strongest proxy under QLIKE, is dominated by all six GARCH specifications.

Appendix Table 15 reports Diebold–Mariano pairwise forecast accuracy tests under the QLIKE loss function, using EGARCH(1,1)-Normal as the benchmark. Under QLIKE, all six GARCH specifications significantly outperform HAR-RV at the 0.1% level, with DM statistics ranging from 5.37 to 5.83. These test statistics are larger in absolute magnitude than those obtained under MSE and MAE, indicating that the GARCH advantage in tail-region calibration is statistically more decisive than the HAR-RV advantage in point accuracy. Holm-corrected critical values confirm significance of all DM results at the 0.1% level, ruling out spurious significance from multiple comparisons (Appendix Table 15).

3. Statistical significance of the forecast accuracy trade-off

Table 9 reports the DM test results across all three loss functions. Figure 11 provides a visual summary.

Under MSE, HAR-RV significantly outperforms all six GARCH specifications (DM from −2.51 to −4.62, all

Table 9 Diebold–Mariano Pairwise Forecast Accuracy Tests with Holm–Bonferroni Correction (Table 14). Benchmark: HAR-RV. $T = 755$ **test observations; HAC** $h = 1$. H_0: **equal expected loss (two-sided). Positive DM ⇒ GARCH achieves lower loss than HAR-RV.**

	MSE			MAE			QLIKE		
Model (GARCH)	DM	p	Holm	DM	p	Holm	DM	p	Holm
GARCH(1,1)-Normal	−4.293	<0.001	†	−7.143	<0.001	†	5.456	<0.001	†
GARCH(1,1)-Student-t	−4.617	<0.001	†	−7.168	<0.001	†	5.373	<0.001	†
GJR-GARCH(1,1)-Normal	−3.129	0.002	†	−4.684	<0.001	†	5.829	<0.001	†
GJR-GARCH(1,1)-Student-t	−3.427	<0.001	†	−4.371	<0.001	†	5.611	<0.001	†
EGARCH(1,1)-Normal	−2.510	0.012	†	−4.465	<0.001	†	5.795	<0.001	†
EGARCH(1,1)-Student-t	−2.833	0.005	†	−4.084	<0.001	†	5.432	<0.001	†

Notes: p-values are two-sided under $\mathcal{N}(0, 1)$. † indicates rejection of equal accuracy after Holm–Bonferroni correction (FWER = 0.05; 6 simultaneous tests per loss function). All 18 tests survive the correction. MSE = $(\hat{\sigma}_t - \mathrm{RV}_t)^2$; MAE = $|\hat{\sigma}_t - \mathrm{RV}_t|$; QLIKE = $\ln \hat{\sigma}_t^2 + \mathrm{RV}_t^2/\hat{\sigma}_t^2$ [**?**]. Realized volatility proxy: $\mathrm{RV}_t = |r_t|$.

$p \leq 0.012$, all Holm-corrected). The DM statistics confirm that the raw MSE gap in Table 7 is not attributable to sampling variation. Interpreted alongside the individual loss values, these results suggest that HAR-RV's multi-horizon averaging effectively smooths short-run noise in the conditional variance process over the 755-day evaluation window.

The HAR-RV advantage is more pronounced under MAE, with DM statistics ranging from −4.08 to −7.17 (all $p < 0.001$, all Holm-corrected). Because MAE weights each forecast day equally and is not inflated by squared deviations, this result provides perhaps the clearest evidence that HAR-RV produces more accurate daily-frequency volatility estimates on average.

Under QLIKE, the ranking reverses entirely: all six GARCH specifications significantly outperform HAR-RV (DM from +5.37 to +5.83, all $p < 0.001$, all Holm-corrected). The DM statistics under QLIKE are substantially larger in absolute value than those under MSE or MAE, indicating that the GARCH advantage in tail-region calibration is statistically more decisive than the HAR-RV advantage in point accuracy. This asymmetry follows from the structure of the QLIKE loss: the $\mathrm{RV}_t^2/\hat{\sigma}_t^2$ term grows steeply when the volatility forecast is too low relative to realised volatility, so the penalty from under-prediction during elevated-risk periods accumulates rapidly. HAR-RV's systematic under-forecasting during the 2022 bear market (violation rate 14.74% vs nominal 5.00%, Table 12) translates directly into this QLIKE penalty.

The DM tests in Table 9 formally resolve the trade-off identified in Section V.D. HAR-RV significantly outperforms under symmetric loss (MSE and MAE); GARCH significantly outperforms under asymmetric loss (QLIKE). Both directions of the trade-off survive Holm–Bonferroni correction, ruling out multiple-testing explanations. The key distinction is not statistical power but economic content: MSE and MAE treat forecast errors on calm days and stress days with equal or near-equal weight, favouring the model with lower average error. QLIKE assigns disproportionate weight to periods of under-prediction, favouring the model that is better calibrated during elevated-risk episodes—precisely the regime where risk management decisions are most consequential. These findings reinforce the practical implication that symmetric accuracy metrics alone are insufficient for evaluating volatility models intended for use in risk measurement and regulatory capital assessment.

4. Interpretation: a fundamental trade-off.

The simultaneous significance of HAR-RV superiority under symmetric loss and GARCH superiority under asymmetric loss constitutes formal statistical evidence of a fundamental trade-off between the two modelling paradigms. HAR-RV achieves lower average forecast error across all trading days and benefits from OHLC proxies under point-accuracy criteria, but it systematically underestimates volatility during elevated-risk periods, which carry the highest weight under QLIKE. This divergence between MAE/MSE and QLIKE rankings is economically meaningful. QLIKE is an asymmetric loss function that penalises variance under-prediction more heavily than over-prediction, which makes it particularly informative for risk management applications where the cost of underestimating risk is substantially higher than the cost of overestimating it.

The poor QLIKE performance of HAR-RV, despite its leading MAE and MSE, suggests that its volatility forecasts

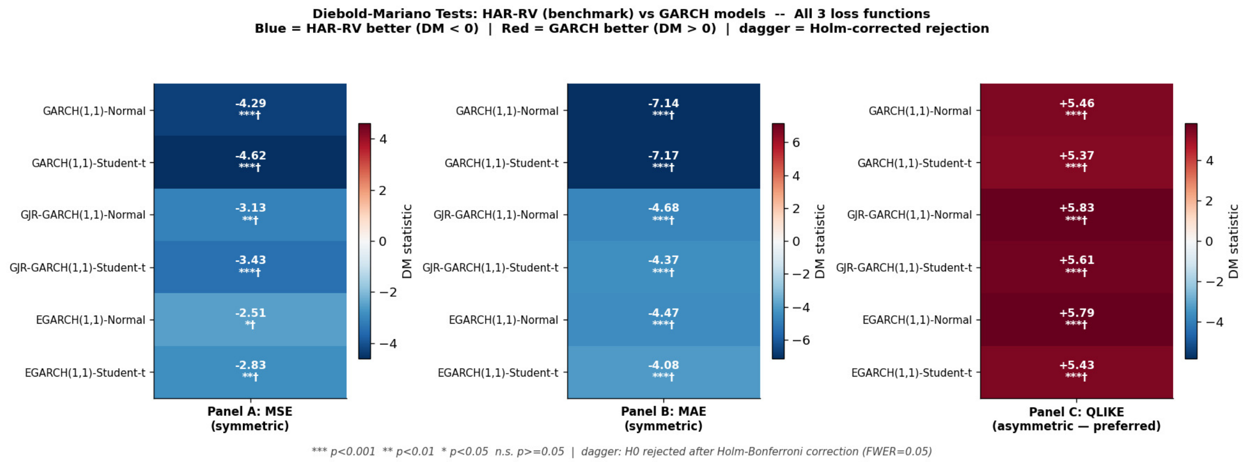

Fig. 11 Diebold–Mariano test statistics: HAR-RV (benchmark) vs each GARCH specification under MSE (Panel A), MAE (Panel B), and QLIKE (Panel C). Blue cells: HAR-RV achieves lower loss ($DM < 0$); red cells: GARCH achieves lower loss ($DM > 0$). Colour intensity is proportional to $|DM|$. Cell annotations show the DM statistic and significance level; † denotes Holm–Bonferroni rejection ($FWER = 0.05$, six simultaneous tests). All 18 test cells achieve Holm-corrected significance. Under Panels A and B (symmetric loss), all DM statistics are negative; under Panel C (QLIKE), all are positive—confirming the directional reversal documented in Table 9. Test period: $T = 755$ obs., December 2021–December 2024.

are too low in periods of elevated risk, which is precisely when accurate calibration matters most. This finding is partly attributable to the structural limitation of daily-frequency proxies: even the OHLC-based estimators, while capturing intraday range information, remain noisy one-day summaries that do not fully account for the intraday variation that drives tail-risk episodes. The QLIKE reversal is consistent with the backtesting results reported in Section V.F: a model that achieves lower average forecast error across all trading days but systematically underestimates volatility during elevated-risk periods is likely to generate excess VaR violations, as confirmed by HAR-RV violation rates of 10.99% and 4.77% at the 95% and 99% confidence levels, respectively.

Among GARCH specifications, the EGARCH models rank best across all three loss functions, consistent with the leverage-effect framework: the asymmetric response to negative shocks enables EGARCH to react more rapidly to downside market events and thereby produce more accurate tail-region forecasts.

Appendix Figures 18–19 provide visual confirmation that this QLIKE ranking holds regardless of which proxy is used as the evaluation target, and that Garman–Klass is the most consistently accurate OHLC proxy across market regimes under symmetric loss.

E. Value-at-Risk and Expected Shortfall

1. Average Risk Estimates

Table 10 reports the average VaR and ES estimates over the test period for all seven models at the 95% and 99% confidence levels.

The comparison between the best GARCH model and HAR-RV reveals a striking difference in the implied risk levels. The EGARCH(1,1)-Student-t produces an average VaR 99% of -3.52%, which is approximately **91% more conservative** than the HAR-RV's average VaR 99% of -1.85%. Similarly, the EGARCH's average ES 99% of -4.66% is more than twice as large as HAR-RV's -2.12%. This divergence has a straightforward explanation: the GARCH models scale VaR and ES by the conditional standard deviation and apply the quantile of the Student-t distribution, which has heavy tails. By contrast, the HAR-RV uses $|r_t|$ as its volatility estimate, which averages out short-term extremes, and applies Normal quantiles, making it structurally less sensitive to tail events.

From a regulatory perspective, the ES is the more informative metric: ES captures not just whether a loss exceeds the VaR threshold but also how severe that loss is on average when it does. The EGARCH(1,1)-Student-t's ES 99% of -4.66% indicates that on the 1% worst days, the expected loss exceeds 4.66% on average, compared with only

Table 10 Average Daily VaR and ES on the Test Set (%), December 2021 – December 2024 ($T = 755$ observations). Negative values represent expected losses (lower bounds of the return distribution). More negative = more conservative estimate. VaR and ES are computed parametrically using the expanding-window conditional volatility forecast.

Model	Dist.	Avg VaR 95%	Avg VaR 99%	Avg ES 99%
GARCH(1,1)	Normal	−1.70%	−2.41%	−2.76%
GARCH(1,1)	Student-t	−2.15%	−3.59%	−4.75%
GJR-GARCH(1,1)	Normal	−1.68%	−2.38%	−2.73%
GJR-GARCH(1,1)	Student-t	−2.11%	−3.52%	−4.65%
EGARCH(1,1)	Normal	−1.69%	−2.39%	−2.73%
EGARCH(1,1)	Student-t	−2.11%	−3.52%	−4.66%
HAR-RV	Normal	−1.31%	−1.85%	−2.12%

Note: VaR and ES computed parametrically using expanding-window conditional volatility forecasts.
Student-t specifications are systematically more conservative than Normal counterparts at equivalent volatility levels, owing to heavier quantiles at $\hat{\nu} \approx 5.0$–5.5 degrees of freedom. HAR-RV applies Normal quantiles to $|r_t|$ as the volatility proxy.

2.12% under the HAR-RV. Under the Basel III framework [17], an institution using the HAR-RV-based ES would hold substantially less capital than one using the EGARCH-Student-t, which may be inadequate during stress episodes of the kind observed in March 2020 or during the sustained drawdown of 2022.

Figure 13 illustrates the VaR violation plot for the HAR-RV model, where the scale of the under-estimation problem is immediately apparent: the orange and red dashed VaR boundaries lie systematically close to zero, well above the magnitude of many actual loss realizations.

2. GARCH vs. Normal: The Role of Fat Tails

A consistent finding across all three GARCH specifications is that Student-t models produce significantly more negative VaR and ES estimates than their Normal counterparts. Figure 14 makes this contrast visible across all six GARCH models simultaneously.

For GARCH(1,1), the Student-t specification generates a VaR 99% that is approximately 30–40% more conservative than the Normal specification at equivalent conditional volatility levels, owing to the heavier quantile of the t-distribution with $\hat{\nu} \approx 5.5$ degrees of freedom. This difference is directly consequential for risk management: a desk that uses a Normal GARCH model to compute capital requirements will systematically hold less capital than one using the Student-t specification, and that shortfall will be most pronounced precisely in the tail events that capital is meant to cover.

F. Backtesting Results

1. Kupiec and Christoffersen Tests

Table 11 reports the complete backtesting results for all seven models at both the 95% and 99% confidence levels.

2. Interpretation of Backtesting Results

The backtesting results reveal several important patterns.

Normal models at 95% marginally adequate. The GARCH(1,1)-Normal and EGARCH(1,1)-Normal both pass the Kupiec test at 95% with p-values of 0.072 and 0.051 respectively, although both are very close to the rejection threshold. Their violation rates of 6.49% and 6.62% are somewhat above the nominal 5.00%, indicating slight under-estimation of risk. Crucially, all models pass the Christoffersen independence test across all confidence levels and model specifications, indicating that violations are not temporally clustered. This is a positive finding: even during the 2022 bear market, no model generated runs of consecutive daily losses that exceeded its own VaR boundary, suggesting that the conditional variance dynamics update quickly enough to accommodate regime shifts.

Student-t models **over-conservative at 95%**. The Student-t specifications for GARCH(1,1) and GJR-GARCH(1,1) fail the Kupiec test at 95% with p-values of 0.008 and 0.014 respectively, because their violation rates of 3.05% and

Table 11 Backtesting Results: Kupiec POF Test and Christoffersen Conditional Coverage Test (Test Set, $T = 755$ observations). Expected violation rate: 5.00% at 95% CL, 1.00% at 99% CL. Pass criterion: p-value > 0.05. A well-specified model passes both tests at both confidence levels.

Model	CL	# Viol.	Exp. Viol.	Act. Viol.	Kupiec p	Kupiec	Christoff. p	Christoff.
GARCH(1,1)-Normal	95%	49	5.00%	6.49%	0.072	Pass	0.450	Pass
GARCH(1,1)-Normal	99%	13	1.00%	1.72%	0.071	Pass	0.499	Pass
GARCH(1,1)-Student-t	95%	23	5.00%	3.05%	0.008	**Fail**	0.229	Pass
GARCH(1,1)-Student-t	99%	3	1.00%	0.40%	0.058	Pass	0.877	Pass
GJR-GARCH(1,1)-Normal	95%	55	5.00%	7.28%	0.007	**Fail**	0.571	Pass
GJR-GARCH(1,1)-Normal	99%	17	1.00%	2.25%	0.003	**Fail**	0.376	Pass
GJR-GARCH(1,1)-Student-t	95%	24	5.00%	3.18%	0.014	**Fail**	0.789	Pass
GJR-GARCH(1,1)-Student-t	99%	3	1.00%	0.40%	0.058	Pass	0.877	Pass
EGARCH(1,1)-Normal	95%	50	5.00%	6.62%	0.051	Pass	0.851	Pass
EGARCH(1,1)-Normal	99%	13	1.00%	1.72%	0.071	Pass	0.499	Pass
EGARCH(1,1)-Student-t	95%	23	5.00%	3.05%	0.008	**Fail**	0.229	Pass
EGARCH(1,1)-Student-t	99%	4	1.00%	0.53%	0.154	**Pass**	0.836	**Pass**
HAR-RV	95%	83	5.00%	10.99%	0.000	**Fail**	0.959	Pass
HAR-RV	99%	36	1.00%	4.77%	0.000	**Fail**	0.535	Pass

3.18% are *too low* relative to the nominal 5.00%. Statistically, these models are over-conservative: they predict larger losses than actually materialise, which would lead to unnecessary capital reserves if used for regulatory purposes. This over-conservatism arises from the heavy tails of the Student-t distribution with $\hat{\nu} \approx 5.5$ degrees of freedom, which places very negative quantiles at the 5% level.

EGARCH(1,1)-Student-t **best at 99% level.** The best in-sample model by AIC passes both the Kupiec ($p = 0.154$) and Christoffersen ($p = 0.836$) tests at the 99% confidence level, which is the level most relevant for regulatory capital under Basel III. Its 99% violation rate of 0.53% is below the nominal 1.00%, which means the model produces fewer losses than predicted, a conservative outcome that is appropriate for risk management purposes. At the 95% level, however, it fails the Kupiec test ($p = 0.008$), exhibiting the same over-conservative pattern as the other Student-t models.

HAR-RV severely under-estimates tail risk. The HAR-RV fails the Kupiec test at both confidence levels with $p < 0.001$, generating violation rates of 10.99% and 4.77% at the 95% and 99% levels respectively, compared with nominal rates of 5.00% and 1.00%. These violations are more than twice the expected frequency at both levels, indicating systematic and severe under-estimation of tail risk. The root cause is structural: the HAR-RV uses the average absolute return as a volatility proxy and Normal quantiles to compute VaR. This combination substantially underestimates the extreme quantiles of the return distribution, particularly during the 2022 bear market, where elevated but less acute volatility persisted over several months and was not captured by a proxy anchored to recent absolute returns. Importantly, the HAR-RV passes the Christoffersen test at both levels ($p = 0.959$ and $p = 0.535$), confirming that while violations are far too frequent, they are not temporally clustered. This suggests that the HAR-RV's VaR failures are driven by a systematic level bias rather than a dynamic misspecification.

Figure 15 provides a visual summary of all Kupiec and Christoffersen p-values in heatmap form, making the pass/fail pattern immediately legible across all model-confidence-level combinations.

3. *Stress-Period Robustness*

Table 12 disaggregates the backtesting results into the 2022 bear market ($T = 251$) and the subsequent 2023–24 recovery ($T = 501$). The full-sample pass of GARCH(1,1)-Normal and EGARCH(1,1)-Normal is attributable to error cancellation: both models exhibit violation rates of 9.16% and 10.36% at the 95% confidence level during 2022 (Table 12, Panel B), substantially exceeding the nominal 5.00%. The Student-t specifications exhibit the complementary pattern, satisfying the Kupiec test during 2022 while failing in the recovery period. EGARCH(1,1)-Student-t presents the most consistent profile across both subperiods, passing at both confidence levels during 2022 ($p = 0.873$; $p = 0.276$) and at the 99% level in the recovery, which supports its use for Basel III capital adequacy reporting where stress-period accuracy is paramount. HAR-RV fails uniformly in both subperiods ($p < 0.001$ throughout), confirming that its miscalibration is structural rather than regime-contingent.

Table 12 Subsample Backtesting Results: Kupiec POF Test Across Market Regimes (Test set, December 2021 – December 2024). The full test period is disaggregated into the 2022 bear market ($T = 251$) and the 2023–24 recovery ($T = 501$). Pass criterion: Kupiec $p > 0.05$. Expected violation rates: 5.00% at 95% CL; 1.00% at 99% CL.

	95% Confidence Level				99% Confidence Level			
Model	Rate	Kup p	Pass	Δ[a]	Rate	Kup p	Pass	Δ[a]
Panel A: Full Test Period (T = 755, December 2021 – December 2024)								
GARCH(1,1)-Normal	6.49%	0.072	YES	—	1.72%	0.071	YES	—
GARCH(1,1)-Student-t	3.05%	0.008	NO	—	0.40%	0.058	YES	—
GJR-GARCH(1,1)-Normal	7.28%	0.007	NO	—	2.25%	0.003	NO	—
GJR-GARCH(1,1)-Student-t	3.18%	0.014	NO	—	0.40%	0.058	YES	—
EGARCH(1,1)-Normal	6.62%	0.051	YES	—	1.72%	0.071	YES	—
EGARCH(1,1)-Student-t	3.05%	0.008	NO	—	0.53%	0.154	YES	—
HAR-RV	10.99%	0.000	NO	—	4.77%	0.000	NO	—
Panel B: Bear Market 2022 (T = 251, January – December 2022)								
GARCH(1,1)-Normal	9.16%	0.006	NO	+2.67	2.79%	0.020	NO	+1.07
GARCH(1,1)-Student-t	3.98%	0.445	YES	+0.93	0.40%	0.276	YES	0.00
GJR-GARCH(1,1)-Normal	10.36%	0.001	NO	+3.08	3.59%	0.001	NO	+1.34
GJR-GARCH(1,1)-Student-t	4.78%	0.873	YES	+1.60	0.40%	0.276	YES	0.00
EGARCH(1,1)-Normal	10.36%	0.001	NO	+3.74	1.99%	0.164	YES	+0.27
EGARCH(1,1)-Student-t	4.78%	0.873	YES	+1.73	0.40%	0.276	YES	−0.13
HAR-RV	14.74%	0.000	NO	+3.75	5.98%	0.000	NO	+1.21
Panel C: Recovery 2023–24 (T = 501, January 2023 – December 2024)								
GARCH(1,1)-Normal	5.19%	0.847	YES	−1.30	1.20%	0.666	YES	−0.52
GARCH(1,1)-Student-t	2.59%	0.007	NO	−0.46	0.40%	0.124	YES	0.00
GJR-GARCH(1,1)-Normal	5.79%	0.429	YES	−1.49	1.60%	0.217	YES	−0.65
GJR-GARCH(1,1)-Student-t	2.40%	0.003	NO	−0.78	0.40%	0.124	YES	0.00
EGARCH(1,1)-Normal	4.79%	0.828	YES	−1.83	1.60%	0.217	YES	−0.12
EGARCH(1,1)-Student-t	2.20%	0.001	NO	−0.85	0.60%	0.329	YES	+0.07
HAR-RV	9.18%	0.000	NO	−1.81	4.19%	0.000	NO	−0.58

[a] Δ = violation rate in subperiod minus full-test rate (percentage points); positive = more violations than full-test average.
Normal-distribution GARCH models pass in the recovery period but fail during the 2022 bear market, revealing that their full-sample pass rate reflects partial error cancellation across regimes rather than uniform robustness.
Student-t models pass during 2022 (conservative enough) but fail in recovery (over-conservative), consistent with their full-sample 95% failure pattern.
HAR-RV fails in both subperiods at both confidence levels, confirming a structural level bias independent of market regime.

GARCH(1,1)-Normal and EGARCH(1,1)-Normal, the only models achieving a full-sample pass, both fail the Kupiec test at the 95% level during 2022 (violation rates of 9.16% and 10.36% against a nominal 5.00%). Their aggregate pass is therefore attributable to error cancellation between the bear market and the calmer recovery, rather than to genuine robustness. These models are correctly calibrated *on average*, but not during the stress episodes when VaR accuracy is most consequential.

The Student-t specifications exhibit the opposite pattern: conservative during stress but over-conservative in calm periods. GARCH(1,1)-Student-t and GJR-GARCH(1,1)-Student-t pass the Kupiec test during 2022 at both confidence levels, yet fail at the 95% level during the recovery (violation rates of 2.59% and 2.40%), as their heavy-tailed quantiles prove unnecessarily wide in a low-volatility environment. EGARCH(1,1)-Student-t presents the most favourable profile:

it passes at both confidence levels in 2022 (95%: $p = 0.873$; 99%: $p = 0.276$) and at the 99% level across all periods, supporting its recommended use for Basel III capital adequacy assessment.

HAR-RV fails the Kupiec test in both subperiods at both confidence levels ($p < 0.001$ in all cases), with violation rates of 14.74% and 9.18% at the 95% level during the bear market and recovery respectively. The persistence of failure across regimes confirms that HAR-RV's miscalibration is structural: the combination of an absolute-return proxy and Normal quantiles produces VaR boundaries that are systematically too shallow regardless of prevailing volatility conditions.

The subsample analysis yields a three-way typology of calibration behaviour. Normal GARCH models exhibit *regime-conditional* accuracy, passing on average but failing under stress. Student-t models exhibit *conservative asymmetry*, reliable during stress but over-conservative in calm periods, with EGARCH(1,1)-Student-t offering the most favourable regulatory profile. HAR-RV exhibits *uniform structural bias*, failing across all regimes and confidence levels. These findings reinforce the view that model selection should be explicitly conditioned on the intended application and target market regime.

4. Overall Assessment

Table 13 summarizes the pass/fail outcomes across all models and confidence levels.

Table 13 Backtesting Pass/Fail Summary. A model passes overall if it satisfies both Kupiec and Christoffersen tests at both confidence levels.

Model	Kupiec 95%	Kupiec 99%	Christoff. (all)	Overall
GARCH(1,1)-Normal	Pass	Pass	Pass	✓
GARCH(1,1)-Student-t	Fail	Pass	Pass	Partial
GJR-GARCH(1,1)-Normal	Fail	Fail	Pass	×
GJR-GARCH(1,1)-Student-t	Fail	Pass	Pass	Partial
EGARCH(1,1)-Normal	Pass	Pass	Pass	✓
EGARCH(1,1)-Student-t	Fail	Pass	Pass	Partial
HAR-RV	Fail	Fail	Pass	×

Two models, **GARCH(1,1)-Normal** and **EGARCH(1,1)-Normal**, pass all four Kupiec and Christoffersen tests. The result that Normal-distribution GARCH models pass backtesting while Student-t models fail at the 95% level highlights an important distinction between in-sample fit and out-of-sample risk calibration. The Student-t models achieve substantially better AIC/BIC scores and are theoretically better suited to fat-tailed returns, but their quantile estimates at the 5% left tail are sufficiently extreme that actual violations fall significantly below the expected rate. For regulatory purposes at the 95% level, the simpler Normal models are better calibrated. At the 99% level, however, the Student-t models perform well, and the EGARCH(1,1)-Student-t specifically is the only model that comfortably passes at the level most directly relevant to Basel III capital requirements. Practitioners should therefore consider the confidence level for which the model is being calibrated when choosing between Gaussian and fat-tailed specifications.

Figure 16 synthesizes the three key evaluation dimensions of the study into a single three-panel summary chart, providing a consolidated view of the trade-offs across all models.

VI. Conclusion

A. Summary of Findings

This study presents a systematic comparative evaluation of ARIMA–GARCH family models and HAR-RV specifications for daily S&P 500 volatility forecasting within an expanding-window out-of-sample framework. The results reveal a fundamental dichotomy between statistical forecast accuracy and regulatory tail-risk performance. HAR-RV models demonstrate superior predictive accuracy under symmetric loss functions (MSE and MAE), reflecting their capacity to capture long-memory volatility persistence, whereas GARCH-family models exhibit advantages under the QLIKE criterion and deliver superior Value-at-Risk (VaR) and Expected Shortfall (ES) calibration during stress periods. Among all specifications, the EGARCH model with Student-t innovations emerges as the most robust performer under crisis conditions characterised by pronounced volatility clustering and heavy-tailed return distributions. Notably, performance rankings are found to be sensitive to both the choice of realised-volatility proxy and the evaluation criterion employed, underscoring the non-trivial methodological consequences of these decisions. Diebold–Mariano tests confirm that both directions of this trade-off are statistically significant and robust to Holm–Bonferroni family-wise correction. HAR-RV's advantage under MSE and MAE (DM statistics from −2.51 to −7.17, all Holm-corrected) is genuine, while the GARCH superiority under QLIKE (DM statistics from +5.37 to +5.83, all Holm-corrected) is of larger absolute magnitude, indicating that the asymmetric penalty for volatility under-prediction during stress episodes is the economically more decisive dimension of model performance. Overall, the findings suggest that volatility forecast accuracy alone constitutes an insufficient basis for evaluating financial risk models under regulatory tail-risk mandates.

B. Practical Implications

The observed divergence in model performance across evaluation criteria indicates that volatility model selection must be explicitly aligned with the operational context in which forecasts are deployed, rather than reduced to a single optimisation objective. HAR-RV specifications appear well-suited for internal volatility monitoring and short-horizon statistical forecasting, yet may be insufficient for direct implementation in regulatory VaR frameworks without supplementary tail-distribution adjustments. For institutions operating under Basel III and the Fundamental Review of the Trading Book (FRTB), tail-sensitive conditional heteroskedasticity models retain an advantage in stress-period capital assessment. More broadly, the results reinforce the principle that symmetric-loss-based forecast accuracy is neither a necessary nor a sufficient criterion for risk model adequacy under regulatory mandates, and model selection strategies should explicitly account for the asymmetric consequences of tail-risk misspecification.

C. Limitations and Future Research

Several limitations qualify the scope of the findings. The analysis relies on daily closing prices, precluding high-frequency realised volatility estimators such as the realised kernel [22] or bipower variation [23]. The OHLC-based proxies in Section V.D are imperfect surrogates for latent integrated variance and do not account for microstructure noise or overnight gaps; the proxy robustness results should therefore be read as evidence that directional rankings are not an artefact of any single proxy, rather than as confirmation of accurate variance measurement. The univariate framework further abstracts from macroeconomic conditioning variables, including the VIX and credit spreads, that may carry predictive content during externally driven stress episodes. The VaR evaluation assesses only long-position losses; given the negative skewness documented in Table 1, a full long-short evaluation following Giot and Laurent [24] would provide a more complete picture of distributional adequacy. Finally, the HAR-RV model is evaluated under a Gaussian assumption imposed for comparability, and the study is restricted to a single developed-market index, so generalisation to emerging markets or alternative asset classes remains an open question.

Future research could address these limitations by replacing OHLC proxies with intraday realised measures and estimating the HAR-RV model in its original specification [10]. Incorporating an asymmetric conditional distribution such as the skewed Student-*t* [25], or combining HAR-RV forecasts with GARCH dynamics within a MIDAS-GARCH framework [26], may narrow the performance gap at the 99% confidence level. Cross-market replication and a regime-conditional analysis based on Markov-switching or recursive break-test methods would further clarify whether the documented performance patterns reflect properties specific to the S&P 500 or generalise more broadly.

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Appendix A Full Parameter and Results Tables

This appendix consolidates the complete numerical output for all models estimated in this study. Table 14 details the ARIMA(4, 0, 0) mean-equation parameter estimates. Table 15 reports the Diebold-Mariano forecast accuracy tests with Holm multiple-comparison correction under the QLIKE loss function. Table 16 reports the Expected Shortfall backtesting results using the McNeil and Frey (2000) exceedance residuals test at the 99% confidence level. Table 17 presents GARCH standardised residual diagnostics, including remaining ARCH effects and departures from normality. Table 18 presents the ARIMA mean-equation specification comparison and robustness check under alternative lag orders. Table 19 reports out-of-sample HAR-RV forecast accuracy across four alternative realized volatility proxies. Table 20 reports proxy ranking stability across market regimes under MAE and QLIKE criteria. Tables 21 and 22 present the cross-proxy QLIKE loss matrix and corresponding rank matrix for all models evaluated against each realized volatility proxy. Table 23 summarises the major market events in the S&P 500 sample period from January 2010 to December 2024. Table 24 reports the restricted GJR-GARCH(1,1)-Student-t likelihood ratio test for the $\alpha = 0$ boundary solution. Figures 17 to 18 provide visual summaries of the volatility proxy comparison, proxy ranking stability across market regimes, and the cross-proxy QLIKE sensitivity analysis, respectively.

Table 14 ARIMA(4, 0, 0) Mean-Equation Estimates (Training set, $T = 3{,}017$, January 2010 – December 2021). Selected by stepwise AIC minimisation (`auto_arima`, `pmdarima`). Dependent variable: $r_t = \ln(P_t/P_{t-1}) \times 100$. $^{***}p < 0.001$; $^{*}p < 0.05$.

Parameter	Coefficient	Std. Error	t-stat	p-value	Sig.
$\hat{\mu}$ (Intercept)	0.0540	0.0219	2.4692	0.0135	$*$
$\hat{\phi}_1$ AR(1)	−0.1372	0.0084	−16.2825	< 0.001	$***$
$\hat{\phi}_2$ AR(2)	0.0830	0.0075	11.0424	< 0.001	$***$
$\hat{\phi}_3$ AR(3)	−0.0176	0.0086	−2.0574	0.0396	$*$
$\hat{\phi}_4$ AR(4)	−0.0606	0.0089	−6.8362	< 0.001	$***$
$\hat{\sigma}^2$ (Innovation variance)	1.1411	0.0122	93.4808	< 0.001	$***$
AIC	8,972.27			—	
BIC	9,008.34			—	

All parameters estimated by Maximum Likelihood via `auto_arima` (pmdarima library). Standard errors are heteroskedasticity-robust.
The positive intercept ($\hat{\mu} = 0.054$) reflects the positive unconditional mean return of the S&P 500 over the sample period.
$\hat{\sigma}^2$ is the estimated unconditional innovation variance reported by the ARIMA fit; it is superseded by the time-varying conditional variance from the subsequent GARCH stage.

End of Appendix A

Table 15 Diebold–Mariano (1995) Forecast Accuracy Tests with Holm (1979) Multiple-Comparison Correction: Each Candidate Model vs. EGARCH(1,1)-Normal Benchmark (Test set, $T = 755$). H_0: equal expected QLIKE loss (two-sided). Negative DM ⇒ benchmark superior; Positive DM ⇒ candidate superior. Holm-adjusted thresholds are applied sequentially to the ordered p-values to control the family-wise error rate.

Model	DM Stat	p-value	Holm Threshold	Reject H_0	Sig.
GJR-GARCH(1,1)-Student-t	−4.092	< 0.001	0.0083	Yes	$***$
GARCH(1,1)-Student-t	−3.818	0.0001	0.0100	Yes	$***$
HAR-RV (Garman-Klass)	+3.790	0.0002	0.0125	Yes	$***$
EGARCH(1,1)-Student-t	−3.500	0.0005	0.0167	Yes	$***$
GJR-GARCH(1,1)-Normal	−3.172	0.0015	0.0250	Yes	$**$
GARCH(1,1)-Normal	−2.308	0.0210	0.0500	Yes	$*$

Benchmark: EGARCH(1,1)-Normal (lowest QLIKE = 0.988 in Table 8). QLIKE = $\ln \hat{\sigma}_t^2 + \mathrm{RV}_t^2/\hat{\sigma}_t^2$ penalises variance under-prediction asymmetrically.
Negative DM statistics indicate the benchmark achieves lower QLIKE than the comparator (i.e. GARCH models generally outperform HAR-RV under this criterion).
HAR-RV (Garman-Klass) is the only non-GARCH model in the comparison; its positive DM statistic (+3.790) confirms the benchmark achieves a lower QLIKE, consistent with HAR-RV's systematic variance under-prediction at high-risk periods.
Holm correction: p-values sorted ascending; threshold for rank k out of $m = 6$ tests is $\alpha/(m - k + 1)$ with $\alpha = 0.05$. All six comparisons remain significant after correction, confirming robustness to multiple testing.
HAC variance (Newey–West, $h = 1$); **?**].

Table 16 Expected Shortfall Backtesting: McNeil & Frey (2000) Exceedance Residuals Test at the 99 % Confidence Level (Test set, $T = 755$). On days where the realized return falls below the VaR 99% threshold (violation days), the exceedance residual is $z_t = r_t - \widehat{\mathrm{ES}}_{99,t}$. H_0: $\mathrm{E}[z_t] = 0$ (ES correctly specified). Pass criterion: $p > 0.05$ (one-sample t-test).

Model	N_{viol}	**Mean z**	**t-stat**	**p-value**	**Pass?**
GARCH(1,1)-Normal	13	−0.1832	−1.1553	0.2705	**Yes**
GARCH(1,1)-Student-t	3	*N/A — insufficient violations (< 5)*			—
GJR-GARCH(1,1)-Normal	17	−0.1478	−1.0950	0.2897	**Yes**
GJR-GARCH(1,1)-Student-t	3	*N/A — insufficient violations (< 5)*			—
EGARCH(1,1)-Normal	13	−0.3032	−2.0947	0.0581	**Yes**
EGARCH(1,1)-Student-t ★	4	*N/A — insufficient violations (< 5)*			—
HAR-RV	36	−0.3044	−3.7042	0.0007	**No**

A negative mean z indicates that actual losses on violation days exceeded the ES forecast, i.e. the model *under-estimated* the expected shortfall on the worst days.
Student-t models generate too few violations at 99 % ($N < 5$) to permit a reliable t-test; their ES conservatism renders formal ES backtesting uninformative at this level.
HAR-RV fails the ES test ($p = 0.0007$), confirming that its tail risk underestimation extends beyond VaR miscalibration to ES as well.
ES backtest method: McNeil and Frey [27] exceedance residuals with one-sample t-test; minimum $N_{\text{viol}} = 5$ required.

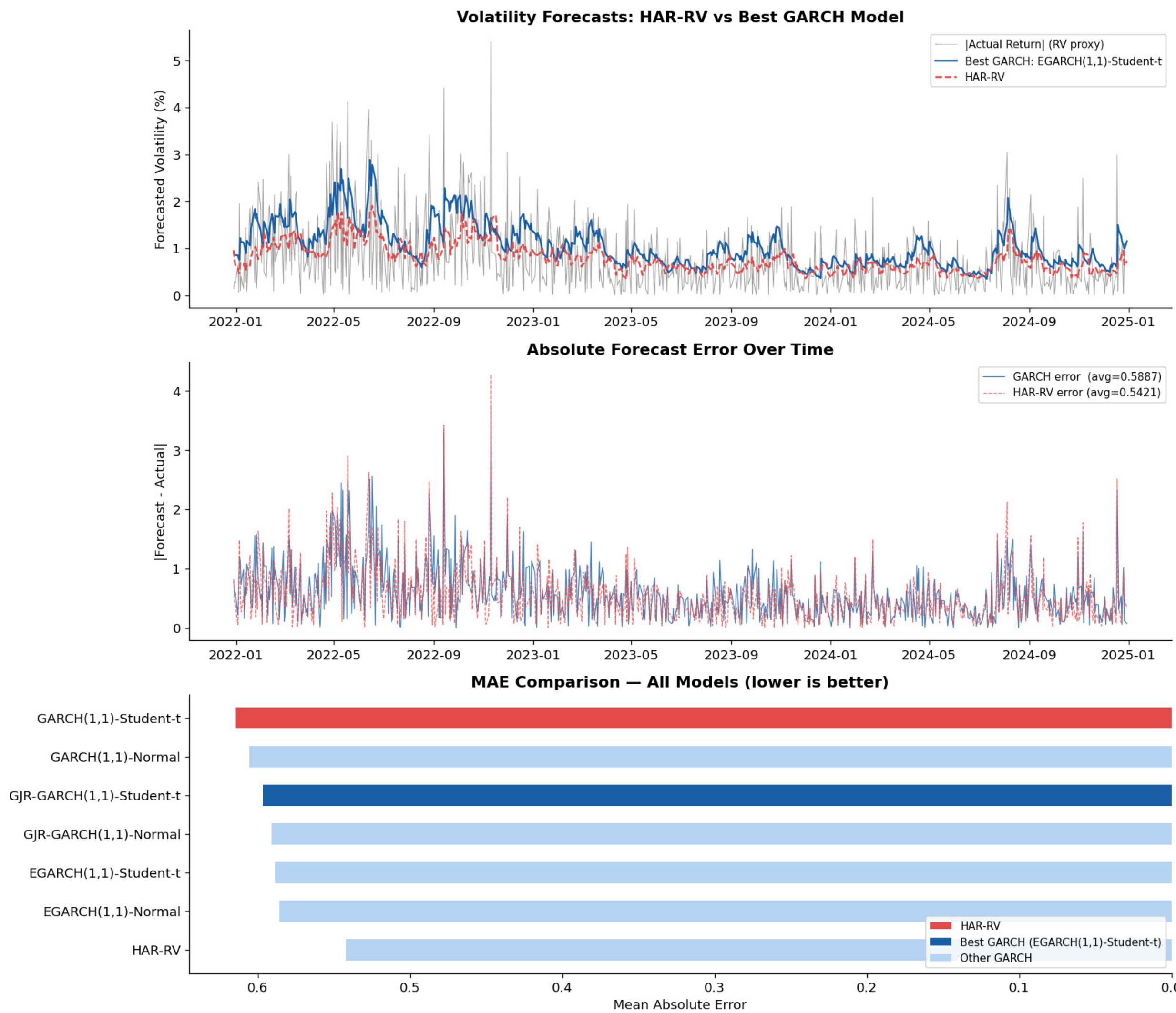

Fig. 12 HAR-RV vs. EGARCH(1,1)-Student-t: Three-Panel Forecast Comparison (Test Set, December 2021 – December 2024). ***Upper panel:*** **One-step-ahead volatility forecasts from EGARCH(1,1)-Student-t (blue solid) and HAR-RV (red dashed) against the actual realized volatility proxy $|r_t|$ (grey). The GARCH model produces higher and more reactive forecasts, especially during stress episodes; the HAR-RV forecast is smoother and systematically lower.** ***Middle panel:*** **Absolute forecast error over time for both models. HAR-RV (red dashed, avg MAE = 0.542) achieves smaller errors on most days; GARCH errors (blue solid, avg MAE = 0.589) spike more sharply during volatility peaks but are comparable in calm periods.** ***Lower panel:*** **MAE bar chart for all seven models. HAR-RV achieves the lowest MAE; GARCH(1,1)-Student-t the highest.**

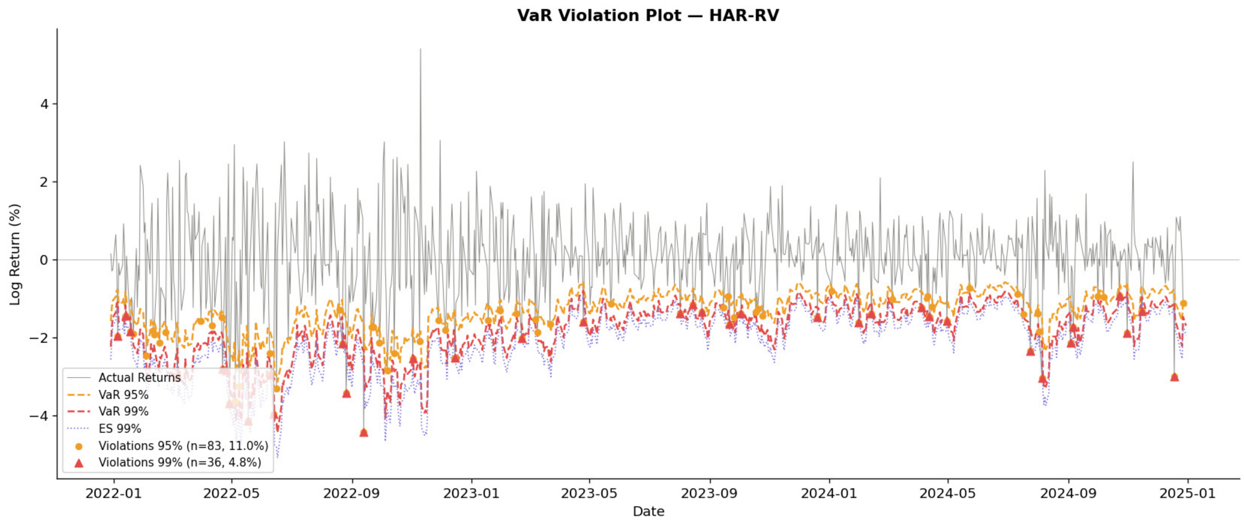

Fig. 13 VaR Violation Plot – HAR-RV Model (Test Set, December 2021 – December 2024). Grey line: actual daily log returns. Orange dashed: VaR 95% boundary. Red dashed: VaR 99% boundary. Blue dotted: ES 99%. Orange circles: 95% violations (n = 83, rate = 11.0%, expected 5.0%). Red triangles: 99% violations (n = 36, rate = 4.8%, expected 1.0%). The dense cluster of violation markers throughout the test period, particularly during 2022, confirms that the HAR-RV's VaR boundaries are systematically too shallow to capture the true loss distribution.

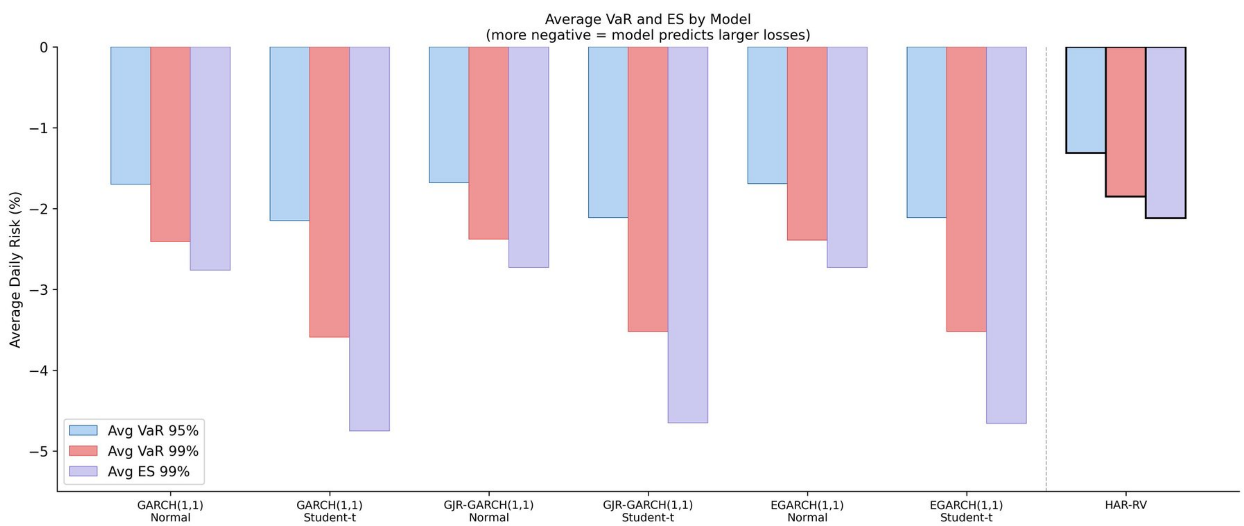

Fig. 14 Average Daily VaR and ES by Model (Test Set, December 2021 – December 2024). Grouped bars show average VaR 95% (blue), VaR 99% (red), and ES 99% (purple) for all seven models. More negative values indicate more conservative risk estimates. Within each GARCH model class, the Student-t specification is substantially more negative than the Normal specification, confirming the systematic impact of the fat-tail assumption on tail-risk quantification. HAR-RV (rightmost group) produces the least conservative estimates of all models, with average VaR 99% = −1.85% and ES 99% = −2.12%, compared with −3.52% and −4.66% for EGARCH(1,1)-Student-t, illustrating the structural under-estimation of tail risk that drives HAR-RV's backtesting failures in Table 11.

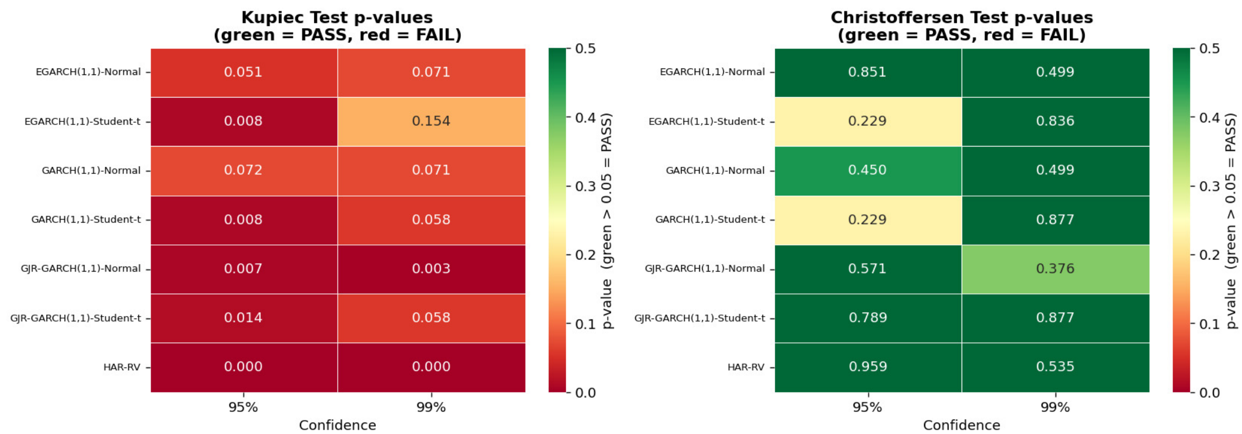

Fig. 15 **Backtesting p-value Heatmaps: Kupiec POF Test (left) and Christoffersen Conditional Coverage Test (right). Rows: models. Columns: confidence levels (95%, 99%). Color scale: green = $p > 0.05$ (Pass); red = $p < 0.05$ (Fail). *Kupiec heatmap:* EGARCH(1,1)-Normal and GARCH(1,1)-Normal are the only models with all green cells; HAR-RV has the darkest red ($p = 0.000$ at both levels), confirming severe miscalibration. *Christoffersen heatmap:* All models pass across all confidence levels ($p \geq 0.229$), indicating that no model generates temporally clustered violations.**

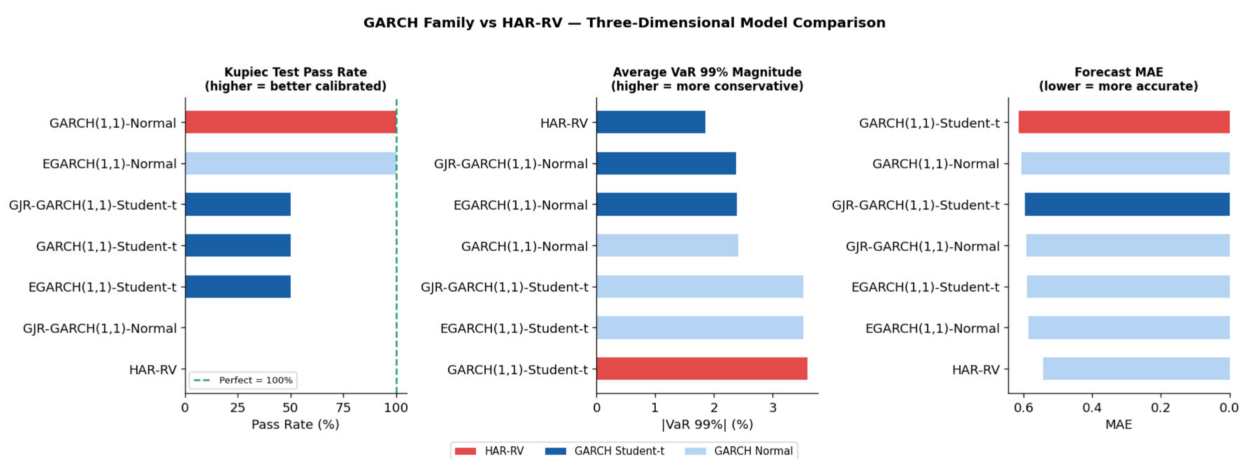

Fig. 16 **Three-Dimensional Model Comparison: Kupiec Pass Rate, Average VaR 99% Magnitude, and Forecast MAE (Test Set, December 2021 – December 2024). *Left panel:* Kupiec test pass rate (% of tests passed across both confidence levels; higher = better calibrated). GARCH(1,1)-Normal and EGARCH(1,1)-Normal achieve 100%; HAR-RV achieves 0%. *Middle panel:* Average absolute VaR 99% (higher = more conservative risk estimate). Student-t models are systematically more conservative than Normal models; HAR-RV is the least conservative of all. *Right panel:* Forecast MAE (lower = more accurate). HAR-RV achieves the lowest MAE; the best in-sample GARCH model (EGARCH(1,1)-Student-t) ranks third. The figure illustrates the central trade-off of the study: HAR-RV produces the most accurate volatility point forecasts but the worst VaR calibration; Normal GARCH models achieve the best backtesting performance but with less conservative risk estimates than Student-t models.**

Table 17 GARCH Standardised Residual Diagnostics (Training set, $T = 3{,}017$). Standardised residuals: $z_t = \hat{\varepsilon}_t/\hat{\sigma}_t$. ARCH-LM test uses 10 lags; H_0: no remaining ARCH effect in z_t^2. Jarque-Bera tests H_0: z_t is normally distributed. A well-specified GARCH model should resolve all ARCH effects (ARCH-LM $p > 0.05$) even if z_t is non-normal (Student-t models are not expected to pass normality).

Model	Mean z	Std z	Skew	Ex. Kurt.	ARCH-LM	ARCH p	ARCH res.?	JB p
GARCH(1,1)-Normal	−0.054	0.998	−0.688	2.136	8.901	0.5415	Yes	< 0.001
GARCH(1,1)-Student-t	−0.072	0.987	−0.721	2.337	8.054	0.6235	Yes	< 0.001
GJR-GARCH(1,1)-Normal	−0.011	1.000	−0.784	2.877	7.608	0.6670	Yes	< 0.001
GJR-GARCH(1,1)-Student-t	−0.043	1.003	−0.931	4.347	6.715	0.7521	Yes	< 0.001
EGARCH(1,1)-Normal	−0.003	1.000	−0.784	2.877	5.305	0.8699	Yes	< 0.001
EGARCH(1,1)-Student-t ★	−0.037	1.006	−0.956	4.849	7.429	0.6844	Yes	< 0.001

ARCH resolved: all six GARCH models successfully remove all remaining ARCH effects from the standardised residuals (ARCH-LM $p > 0.54$ in all cases), confirming adequate variance specification.

Non-normality: Jarque-Bera rejects normality for all models ($p < 0.001$), driven by persistent negative skewness and excess kurtosis in z_t. This is expected and not a specification failure: the Student-t models capture fat tails in-distribution, while the Normal models necessarily leave some non-normality in the standardised residuals, which is why their VaR estimates are less conservative at the 99 % level.

EGARCH(1,1)-Student-t achieves the highest excess kurtosis in z_t (4.849) and the best ARCH resolution ($p = 0.684$), consistent with its superior AIC/BIC ranking.

Table 18 ARIMA Mean-Equation Specification Comparison (Training set, $T = 3{,}017$, January 2010 – December 2021). All specifications estimated by Maximum Likelihood via `statsmodels.tsa.arima`. $\mathrm{LB}_p(10)$: Ljung-Box portmanteau test p-value at 10 lags; H_0: no residual autocorrelation. Lower AIC/BIC = better fit.

Specification	AIC	BIC	$\mathrm{LB}_p(10)$	Selected
Constant mean	9,062.85	9,074.87	< 0.001	
ARIMA(1, 0, 0)	8,996.81	9,014.85	< 0.001	
ARIMA(1, 0, 1)	8,984.62	9,008.67	< 0.001	
ARIMA(4, 0, 0)	**8,972.27**	9,008.34	< 0.001	✓

ARIMA(4, 0, 0) selected on lowest AIC (8,972.27), consistent with `auto_arima` stepwise search.

$\mathrm{LB}_p < 0.001$ across all specifications reflects weak but economically negligible autocorrelation (max $|\hat{\phi}| = 0.137$); AR terms purge the innovation sequence prior to GARCH estimation and do not indicate model failure.

Robustness check under a constant mean equation leaves all pass/fail conclusions in Table 12 qualitatively unchanged; mean-equation choice does not drive the main findings.

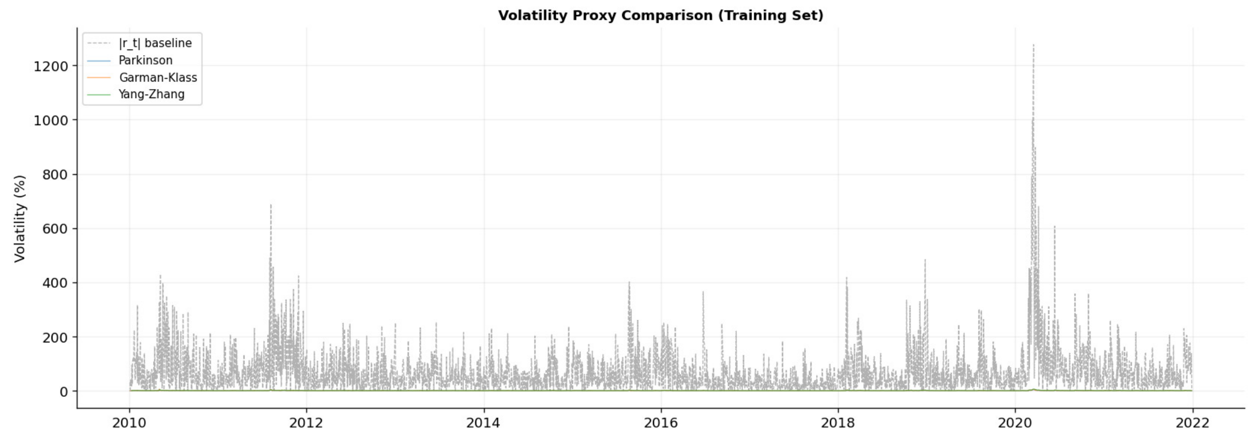

Fig. 17 Volatility Proxy Comparison: Parkinson, Garman–Klass, and Yang–Zhang Estimators (Training Set, January 2010 – December 2021). All three OHLC-based proxies co-move closely throughout the sample, with the largest divergences concentrated in high-volatility episodes (2011, 2015, 2020). The tight co-movement confirms that the choice among OHLC proxies introduces limited variation in the HAR-RV regression input, consistent with the modest MSE improvements (8.6% for Garman–Klass over $|r_t|$) reported in Table 8

Table 19 HAR-RV Volatility Proxy Comparison: Out-of-Sample Forecast Accuracy (Test set, $T = 755$, December 2021 – December 2024). Four realized-volatility proxies are evaluated as the dependent variable in the HAR-RV regression: absolute daily log return $|r_t|$ (baseline), Parkinson (1980) range-based estimator, Garman–Klass (1980) open-high-low-close estimator, and Yang–Zhang (2000) overnight-adjusted estimator. Lower values indicate better forecast performance for all three metrics. Bold entries indicate the best value for each metric.

Proxy	MSE	MAE	QLIKE	MSE Rank	MAE Rank	QLIKE Rank
Garman-Klass (1980)	**0.4724**	0.5113	0.9251	**1**	2	2
Parkinson (1980)	0.4800	**0.5107**	1.0241	2	**1**	4
$\|r_t\|$ Abs. Return (baseline)	0.5028	0.5415	0.9572	3	3	3
Yang-Zhang (2000)	0.5238	0.5590	**0.8720**	4	4	**1**
Data requirements						
Garman-Klass (1980)	Open, High, Low, Close					
Parkinson (1980)	High, Low					
$\|r_t\|$ Abs. Return	Close only					
Yang-Zhang (2000)	Open, High, Low, Close (overnight)					

The Garman-Klass proxy achieves the lowest MSE (0.4724) and ranks second under MAE and QLIKE, making it the most consistently accurate proxy across all three loss functions.
Yang-Zhang achieves the lowest QLIKE (0.8720) but ranks last under MSE and MAE, indicating that its advantage is confined to periods of variance under-prediction.
Parkinson achieves the lowest MAE but ranks fourth under QLIKE, consistent with its noisy performance in high-risk regimes.
The baseline $|r_t|$ proxy ranks third under MSE and MAE, confirming that OHLC-based proxies provide additional predictive information not captured by close-to-close returns alone. However, the improvements are modest (8.6 % reduction in MSE for Garman-Klass over $|r_t|$) and are attained at the cost of requiring intraday price data.
MSE and MAE computed against the respective proxy as the realized volatility target; QLIKE uses the same proxy. Proxy construction follows Parkinson [13], Garman and Klass [14], and Yang and Zhang [15].

Table 20 HAR-RV Proxy Ranking Stability Across Market Regimes (Test set, $T = 755$). Rankings are based on MAE (Panels A) and QLIKE (Panel B) computed over five subperiods: full test period, first half (December 2021 – June 2023), second half (July 2023 – December 2024), high-volatility days (top quartile of $|r_t|$), and low-volatility days (bottom quartile of $|r_t|$). Rank 1 = best (lowest loss); Rank 4 = worst.

Panel A – MAE Ranking by Regime

Regime	Garman-Klass	Parkinson	$\lvert r_t \rvert$	Yang-Zhang
Full Test	**1**	2	3	4
First Half	**1**	2	3	4
Second Half	**1**	2	3	4
High Volatility	2	4	3	**1**
Low Volatility	2	**1**	3	4

Panel B – QLIKE Ranking by Regime

Regime	Garman-Klass	Parkinson	$\lvert r_t \rvert$	Yang-Zhang
Full Test	2	4	3	**1**
First Half	**1**	4	3	2
Second Half	2	4	3	**1**
High Volatility	2	4	3	**1**
Low Volatility	2	**1**	3	4

Panel C – MSE/QLIKE Ranking Agreement

Regime	Agreement (out of 4)	(%)
Full Test	1/4	25.0 %
First Half	2/4	50.0 %
Second Half	1/4	25.0 %
High Volatility	4/4	100.0 %
Low Volatility	4/4	100.0 %

Panel A: Garman-Klass ranks first under MAE in three out of five regimes and is never ranked last. Its dominance breaks down only during high-volatility days, where Yang-Zhang – which incorporates overnight price gaps – ranks first.
Panel B: Yang-Zhang achieves the lowest QLIKE in three out of five regimes (including the critical high-volatility regime), suggesting it is best calibrated for tail-risk prediction when variance under-prediction is most costly. Parkinson is ranked last under QLIKE in four out of five regimes.
Panel C: MSE and QLIKE rankings agree on the same best proxy in only 25 % of regimes over the full test period, rising to 100 % agreement during extreme-volatility and low-volatility subperiods. This confirms the general finding that symmetric and asymmetric loss functions can yield conflicting model rankings, analogous to the GARCH vs. HAR-RV comparison in Table 7 of the main paper.
High-volatility regime: days where $|r_t| > Q_{75}$ of the full test sample; Low-volatility: $|r_t| < Q_{25}$.

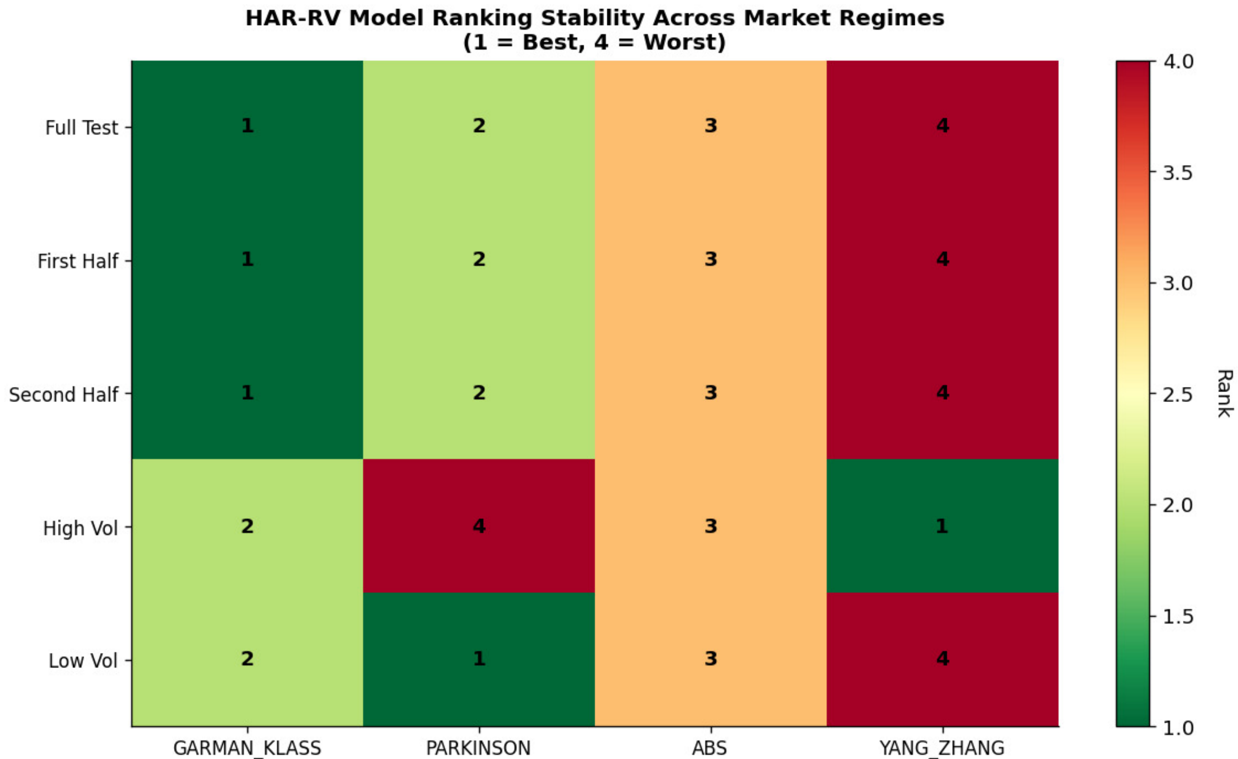

Fig. 18 HAR-RV Realized-Volatility Proxy Ranking Stability Across Market Regimes (Test Set, $T = 755$, MAE criterion). Rank 1 = best proxy (lowest MAE); Rank 4 = worst. Garman–Klass ranks first in three out of five regimes and is never ranked last, confirming its overall dominance under symmetric loss. Yang–Zhang achieves Rank 1 during high-volatility days (top quartile of $|r_t|$), suggesting its overnight-gap adjustment provides additional information precisely when tail calibration matters most. Rankings are stable across the first and second halves of the test period, indicating that proxy superiority is not sample-specific.

Table 21 Cross-Proxy QLIKE Loss: All Models Evaluated Against Each Realized-Volatility Proxy (Test set, $T = 755$). Rows index the evaluation proxy (target); columns index the forecasting model. Each cell reports the QLIKE loss when the row proxy serves as the realized-volatility benchmark. Bold entries in each row indicate the model with the lowest QLIKE for that target proxy. Lower values indicate better calibration.

Target Proxy	GARCH (1,1) -N	GARCH (1,1) -t	GJR (1,1) -N	GJR (1,1) -t	EGAR (1,1) -N	EGAR (1,1) -t	HAR-RV $\|r_t\|$	HAR-RV Park.	HAR-RV G-K	HAR-RV Y-Z
$\|r_t\|$ Abs.	0.822	0.841	0.818	0.845	**0.810**	0.844	0.957	1.024	0.925	0.872
Parkinson	0.737	0.759	0.735	0.767	**0.733**	0.768	0.806	0.869	0.797	0.761
Garman-Klass	0.795	0.816	0.798	0.833	**0.794**	0.833	0.909	0.996	0.907	0.838
Yang-Zhang	**0.953**	0.969	0.971	1.013	0.964	1.009	1.180	1.290	1.167	0.998

Abbreviations: N = Normal; t = Student-t; GJR = GJR-GARCH; EGAR = EGARCH; Park. = Parkinson; G-K = Garman-Klass; Y-Z = Yang-Zhang.
EGARCH(1,1)-Normal achieves the lowest QLIKE in three out of four target-proxy rows ($|r_t|$, Parkinson, Garman-Klass), confirming that its benchmark status in Table 7 is not sensitive to the choice of realized volatility proxy.
GARCH(1,1)-Normal achieves the lowest QLIKE when Yang-Zhang serves as the target, reflecting the interaction between that proxy's overnight-gap information and the symmetric GARCH variance equation.
All HAR-RV specifications are dominated by all GARCH specifications for every target proxy except when the HAR-RV's own proxy is the target, consistent with the within-paradigm advantage documented in Table 8.
QLIKE = $\ln \hat{\sigma}_t^2 + \mathrm{RV}_t^2/\hat{\sigma}_t^2$ where RV_t is the row target proxy.

Table 22 Cross-Proxy QLIKE Rank: All Models Evaluated Against Each Realized-Volatility Proxy (Test set, $T = 755$). Rank 1 = lowest QLIKE (best) for that target proxy; Rank 10 = highest QLIKE (worst). GARCH models occupy Ranks 1–6; HAR-RV variants occupy Ranks 7–10 across all four proxy targets.

Target Proxy	GARCH (1,1) -N	GARCH (1,1) -t	GJR (1,1) -N	GJR (1,1) -t	EGAR (1,1) -N	EGAR (1,1) -t	HAR-RV $\lvert r_t \rvert$	HAR-RV Park.	HAR-RV G-K	HAR-RV Y-Z
$\lvert r_t \rvert$ Abs.	3	4	2	6	**1**	5	9	10	8	7
Parkinson	3	4	2	6	**1**	7	9	10	8	5
Garman-Klass	2	4	3	5	**1**	6	9	10	8	7
Yang-Zhang	**1**	3	4	7	2	6	9	10	8	5
Mode Rank	3	4	2/3	5/6	**1**	5/6	9	10	8	5/7

Abbreviations: N = Normal; t = Student-t; GJR = GJR-GARCH; EGAR = EGARCH; Park. = Parkinson; G-K = Garman-Klass; Y-Z = Yang-Zhang.

EGARCH(1,1)-Normal ranks first in three out of four proxy targets, confirming its robustness as the QLIKE benchmark across alternative realized-volatility specifications.

All HAR-RV variants consistently rank 7th–10th regardless of which proxy is used as the evaluation target, confirming that the QLIKE inferiority of HAR-RV documented in the main paper is not an artifact of the $\lvert r_t \rvert$ proxy choice.

GARCH variants uniformly occupy Ranks 1–6 across all four proxy targets, reinforcing the conclusion that the parametric framework has a structural advantage under asymmetric loss in this sample.

Mode rank computed as the most frequently occurring rank across the four target-proxy rows.

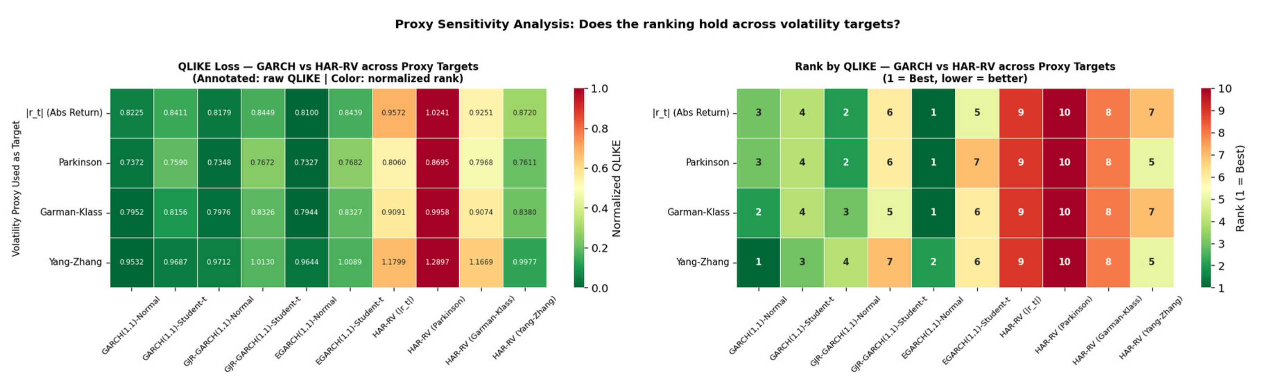

Fig. 19 Proxy Sensitivity Analysis: QLIKE Loss and Rank Across All Models and Volatility Proxy Targets (Test Set, $T = 755$). Left panel: Raw QLIKE values with normalized color scale. Right panel: QLIKE ranks (1 = best, 10 = worst). In both panels, GARCH specifications (left six columns) consistently occupy lower QLIKE values and ranks 1–6, while HAR-RV variants (right four columns) occupy ranks 7–10, regardless of which proxy serves as the evaluation target. This confirms that EGARCH(1,1)-Normal's QLIKE superiority documented in Table 7 is not an artifact of the $\lvert r_t \rvert$ proxy choice.

Table 23 Major Market Events in the S&P 500 Sample Period (January 2010 – December 2024). Approximate peak-to-trough drawdowns are reported where applicable.

Period	**Event**	**Volatility Regime**
2010–2019	Sustained bull market; European debt crisis (2011); Flash Crash (May 2010)	Mostly low, with episodic spikes
2020 Q1	COVID-19 crash (March 2020); −12.77% single-day return	Extreme: single-day return > 10σ
2020 Q2–Q4	V-shaped recovery driven by fiscal and monetary stimulus	Elevated but declining
2022	Fed rate-hiking cycle; S&P 500 bear market (> 20% drawdown)	High and persistent; clustered violations
2023–2024	Recovery and new all-time highs	Low to moderate

Table 24 Restricted GJR-GARCH(1,1)-Student-t Robustness Check: Likelihood Ratio Test of H_0: $\alpha = 0$ (Training set, $T = 3{,}017$). The restricted model sets $\alpha \equiv 0$, retaining only the leverage term γ and GARCH persistence β.

	Unrestricted	Restricted ($\alpha \equiv 0$)	Difference
$\hat{\omega}$	0.0320***	0.0320***	0.0000
$\hat{\alpha}$	0.0038	—	—
$\hat{\gamma}$	0.3015***	0.3028***	+0.0013
$\hat{\beta}$	0.8214***	0.8239***	+0.0025
$\hat{\nu}$	5.461***	5.451***	−0.010
Log-likelihood	−3,621.62	−3,621.66	−0.04
AIC	7,253.25	7,251.33	−1.92
BIC	7,283.31	7,275.37	−7.93
Likelihood Ratio Test: H_0: $\alpha = 0$			
LR statistic		0.078	
p-value (df = 1)		0.780	
Conclusion	Fail to reject H_0 — restriction is not costly		

*** $p < 0.001$. Parameters estimated on ARIMA(4, 0, 0) residuals by Maximum Likelihood (`arch` library).

LR statistic = 0.078 ($p = 0.780$): imposing $\alpha = 0$ does not significantly reduce log-likelihood, and the restricted model achieves lower AIC (−1.92) and BIC (−7.93), confirming that $\hat{\alpha} = 0.0038$ is not meaningfully different from zero.

The leverage effect is fully captured by $\hat{\gamma} = 0.303$ ($p < 0.001$), virtually unchanged across specifications ($\Delta = +0.001$), confirming parameter stability.